

AWS SimuLearn: Creación de un Asistente Inteligente con IA (Amazon Bedrock)

Parámetro	Detalle
Dificultad	Avanzada
Tiempo Estimado	90 Minutos
Servicios Principales	Amazon Bedrock (Agents & Knowledge Bases), Amazon OpenSearch Serverless, AWS Lambda, Amazon S3, Amazon DynamoDB

1. Resumen y Objetivos

Este laboratorio práctico se centra en la implementación de una solución de **IA Generativa** completa utilizando una arquitectura de **Generación Aumentada por Recuperación (RAG)**. Configurarás un asistente inteligente capaz de consultar documentos privados y ejecutar acciones transaccionales en tiempo real.

Los objetivos principales son:

- Configurar una **Knowledge Base** en Amazon Bedrock vinculada a un almacén de vectores en OpenSearch Serverless.
- Orquestar un **Amazon Bedrock Agent** utilizando modelos de lenguaje avanzados (LLM).
- Implementar **Action Groups** mediante funciones AWS Lambda para persistir datos en Amazon DynamoDB.
- Validar el flujo end-to-end: desde la consulta del usuario hasta la ejecución de la lógica de negocio.

2. Análisis del Escenario

El cliente requiere un asistente de Recursos Humanos (HR) que reduzca la carga operativa del departamento. El asistente debe cumplir dos funciones críticas:

1. **Consulta de información:** Responder preguntas basadas en manuales de empleados y políticas de compensación (Datos no estructurados).
2. **Ejecución de procesos:** Procesar solicitudes de vacaciones de empleados e insertarlas en la base de datos de la empresa (Acciones programáticas).

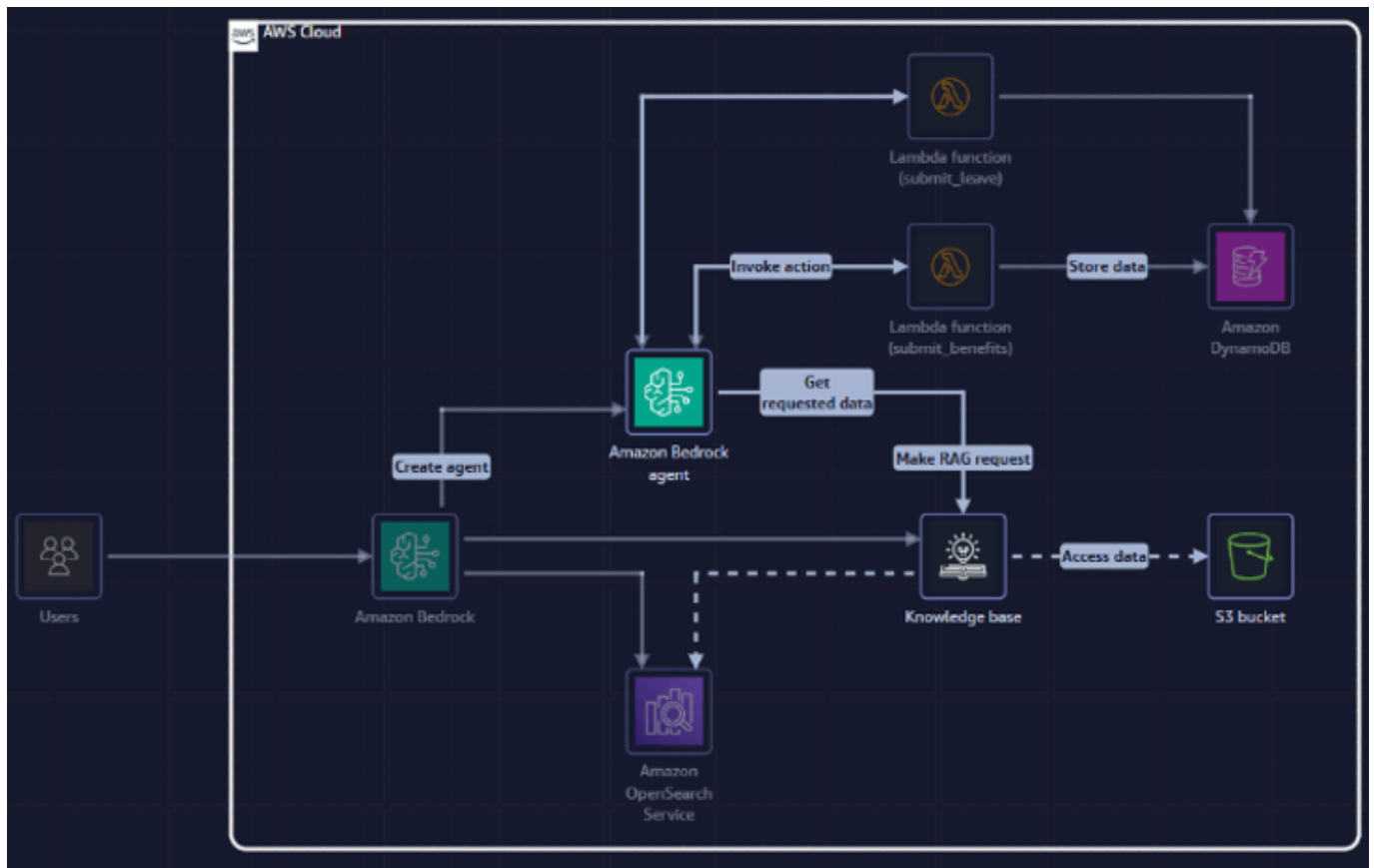
Como Ingeniero de AWS, implementarás Amazon Bedrock como el motor central de inferencia, asegurando que el modelo solo acceda a la información autorizada en el bucket de S3.

3. Arquitectura

La solución sigue un flujo de trabajo moderno de IA:

1. **Almacenamiento:** Documentos en **S3** y registros en **DynamoDB**.

2. **Indexación: OpenSearch Serverless** almacena los *embeddings* (vectores) de los documentos para búsquedas semánticas.
3. **Capa de IA: Amazon Bedrock** gestiona la base de conocimientos y el agente.
4. **Capa de Computación: Lambda** actúa como el puente entre el agente y la base de datos para ejecutar acciones.

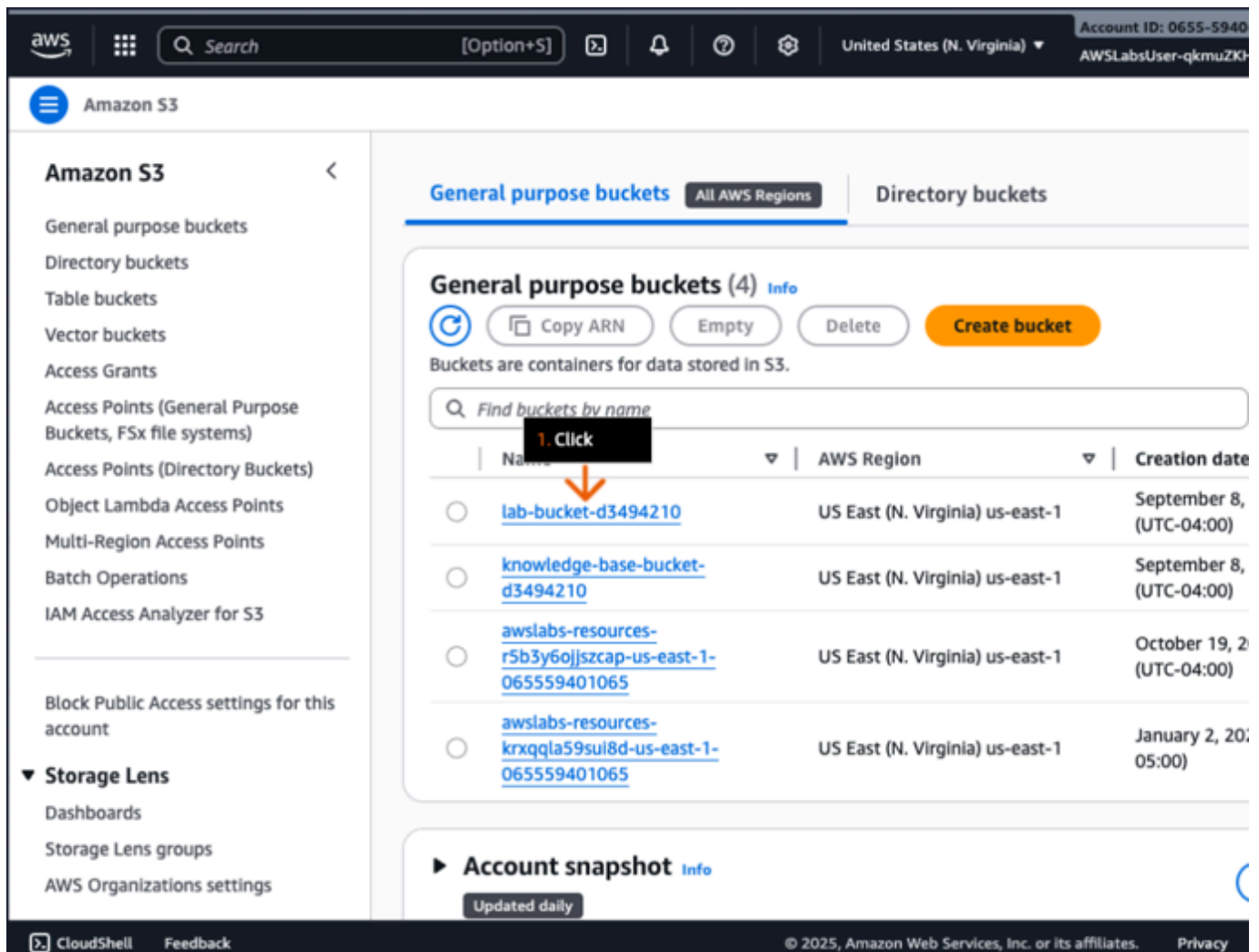
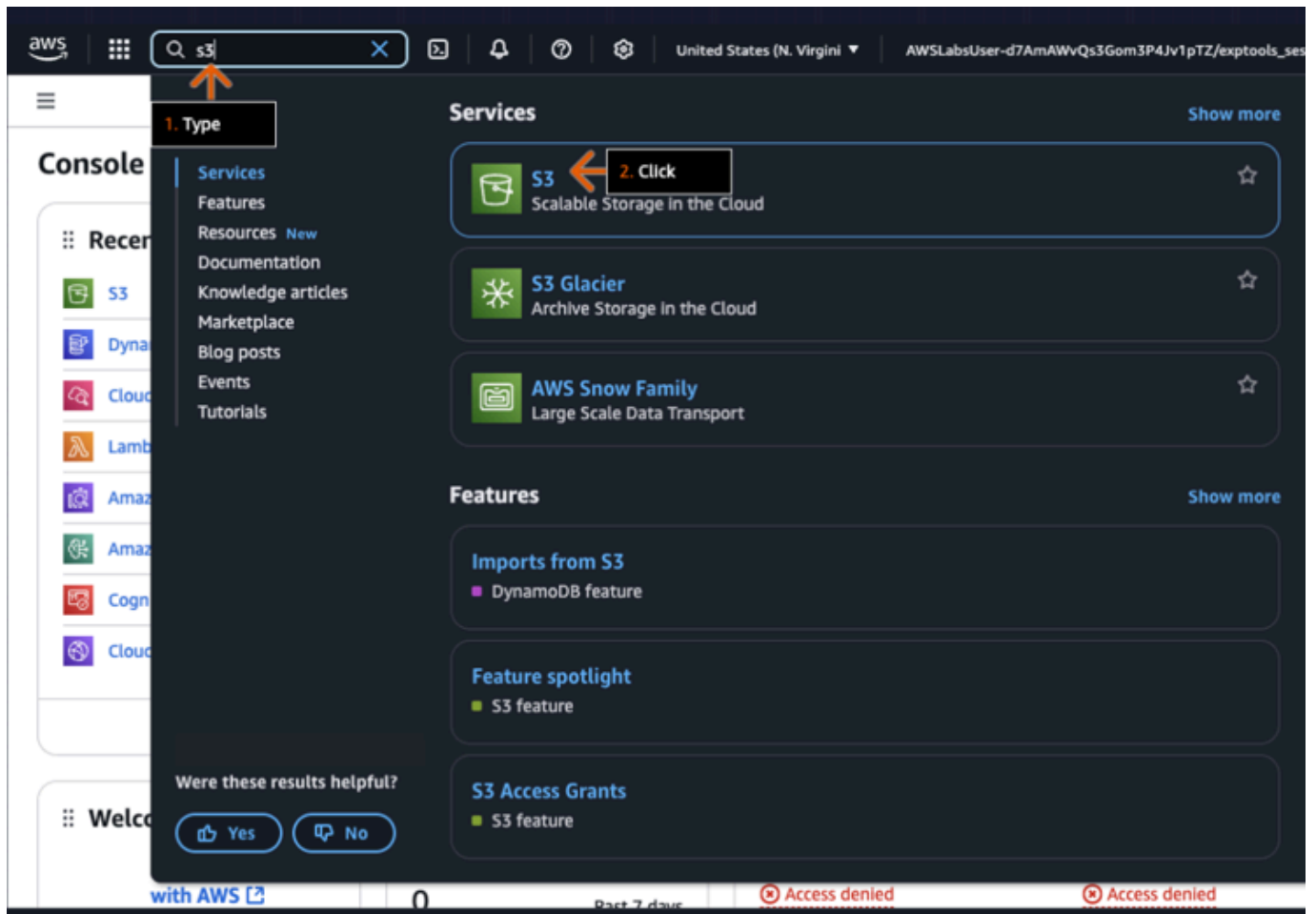


4. Desarrollo de las Tareas Paso a Paso

Tarea 1: Recopilación de Recursos y Prompts

Antes de configurar la IA, debes obtener las instrucciones y los datos base.

1. **Accede a Amazon S3:** Navega a la consola de S3 y localiza el bucket `lab-bucket-d3494210`.



2. **Obtén el Prompt del Agente:** Abre el archivo `agent-prompt.txt` y **copia** su contenido. Este archivo define la personalidad y responsabilidades del asistente de HR.

aws

[Option+S]

United States (N. Virginia)

Account ID: 0655-5940-1065

AWSLabsUser-qkmuZKHWRG8XgJLRHdvTgV/exptoo...

Amazon S3

Buckets

lab-bucket-d3494210

lab-bucket-d3494210

Info

Objects

Metadata

Properties

Permissions

Metrics

Management

Ac

Objects (1/1)

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 Explorer](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

1

1. Choose

agent-prompt.txt

txt

September 8, 2025, 18:28:26 (UTC-04:00)

You are a professional HR Assistant responsible for helping employees with HR-related policy questions. Follow these guidelines:

1. Primary Responsibilities:

- Answer questions about HR policies using all knowledge bases available

- Maintain professional and friendly tone

- Protect confidential information

2. When handling HR policy questions:

- Include relevant policy references only from available knowledge bases

- If the information is not in the knowledge bases, recommend consulting HR department

3. Limitations:

- Defer sensitive matters to HR department

4. Response Format:

- Be concise and clear

- Confirm understanding before proceeding

- Provide next steps when applicable

Always maintain professionalism and confidentiality in all interactions.

1. Copy

3. **Revisa los manuales:** En el bucket `knowledge-base-bucket-...`, confirma la existencia de los archivos `employee_handbook.txt` y `compensation_handbook.txt`. Estos alimentarán la Knowledge

Base.

Amazon S3 > Buckets **1. Return**

General purpose buckets All AWS Regions Directory buckets

General purpose buckets (4) Info

Copy ARN Empty Delete Create bucket

Buckets are containers for data stored in S3.

Find buckets by name

Name	AWS Region
lab-bucket-d3494210	US East (N. Virginia) us-east-1
knowledge-base-bucket-d3494210 2. Click	US East (N. Virginia) us-east-1
awslabs-resources-r5b3y6ojjszcap-us-east-1-065559401065	US East (N. Virginia) us-east-1
awslabs-resources-krxqqla59sui8d-us-east-1-065559401065	US East (N. Virginia) us-east-1

Account snapshot Info View dashboard

Updated daily

Storage Lens provides visibility into storage usage and activity trends.

External access summary - new Info

Updated daily

Amazon S3 > Buckets > knowledge-base-bucket-d3494210

knowledge-base-bucket-d3494210 Info

Objects Metadata Properties Permissions Metrics Management Actions

Objects (2)

Copy S3 URI Copy URL Download Open Delete

Actions Create folder Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

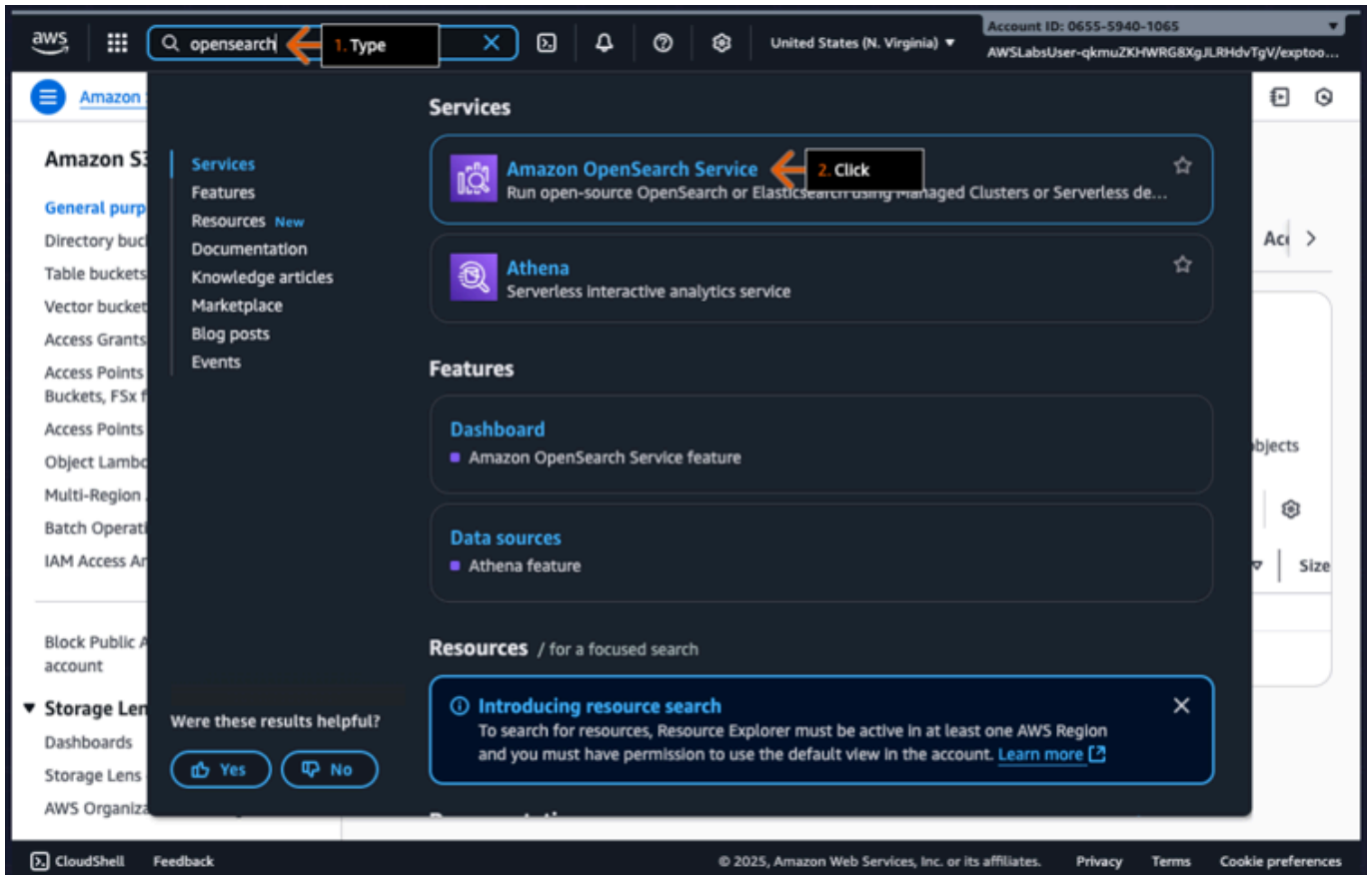
Name	Type	Last modified	Size
compensation_handbook.txt	txt	September 8, 2025, 18:28:26 (UTC-04:00)	
employee_handbook.txt	txt	September 8, 2025, 18:28:26 (UTC-04:00)	

1. Review

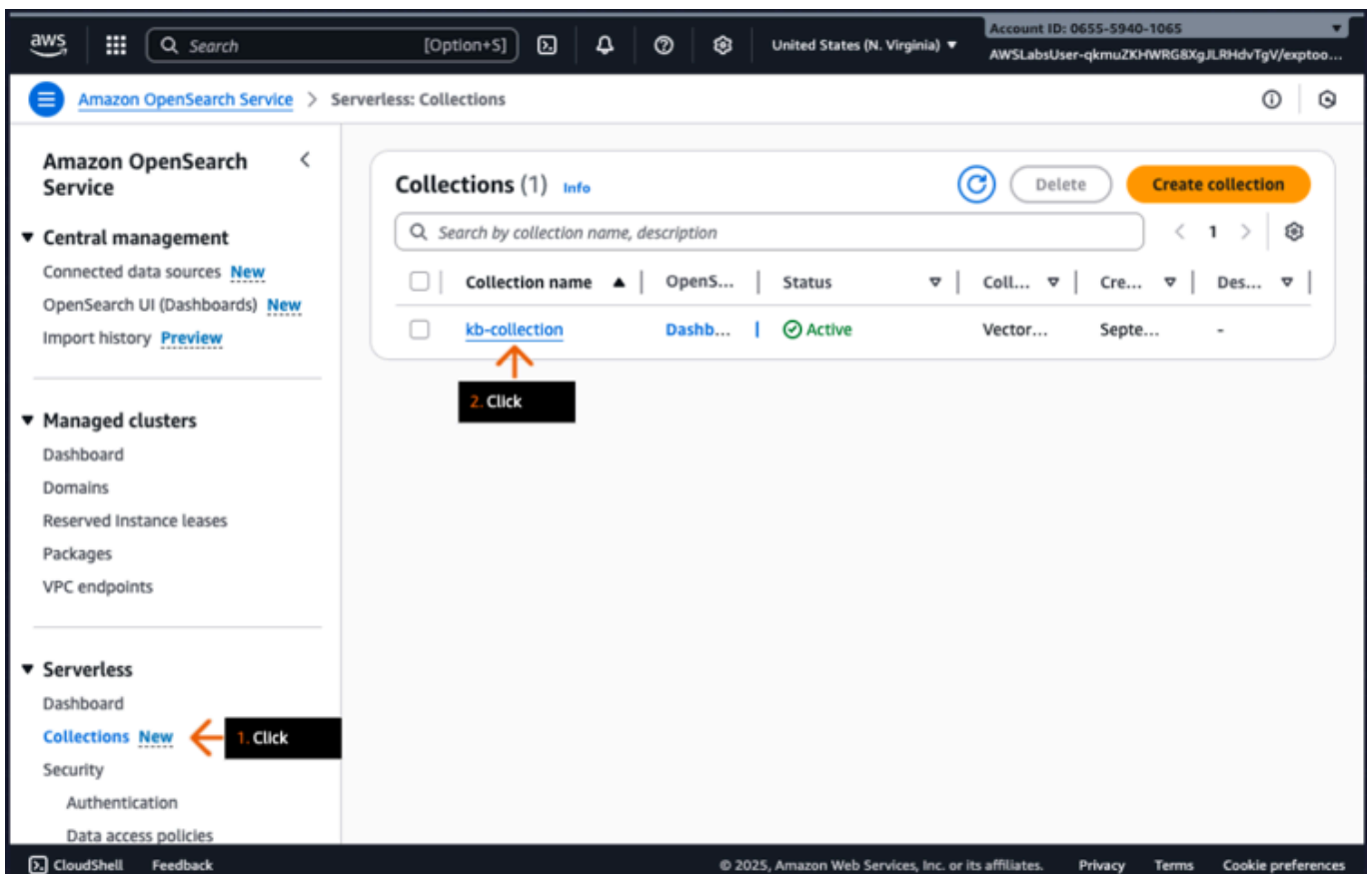
Tarea 2: Configuración del Motor de Búsqueda Vectorial

Amazon Bedrock requiere un lugar donde buscar información de forma semántica.

1. Navega a Amazon OpenSearch Service: Ve a la sección **Serverless** > **Collections**.



2. Identifica la Colección: Selecciona **kb-collection** y copia su **ARN**. Lo necesitarás para vincular la Knowledge Base.



Amazon OpenSearch Service > Serverless: Collections > kb-collection

Overview | Monitor | **Indexes** | Tags

General information

Status: Active

Collection type: Vectorsearch

Indexes: 1

Collection description: -

Total size: 0B

Deployment type: Enabled redundancy

Creation date: September 8, 2025, 18:26 (UTC-04:00)

Collection ARN: [arn:aws:aoss:us-east-1:065559401065:collection/v3ebitydm2r9brh7su21](#)

OpenSearch endpoint (FIPS protection enabled): [https://v3ebitydm2r9brh7su21.us-east-1.aoss-fips.amazonaws.com](#)

OpenSearch Dashboards URL (FIPS protection enabled): [https://dashboards.us-east-1.aoss-fips.amazonaws.com/_login/?collectionid=v3ebitydm2r9brh7su21](#)

OpenSearch endpoint: [https://v3ebitydm2r9brh7su21.us-east-1.aoss-fips.amazonaws.com](#)

OpenSearch Dashboards URL: [https://dashboards.us-east-1.aoss-fips.amazonaws.com/_login/?collectionid=v3ebitydm2r9brh7su21](#)

3. **Verifica el Índice:** Confirma que existe un índice llamado **bedrock-knowledge-base-index** con dimensiones configuradas para vectores (usualmente 1024 o 1536).

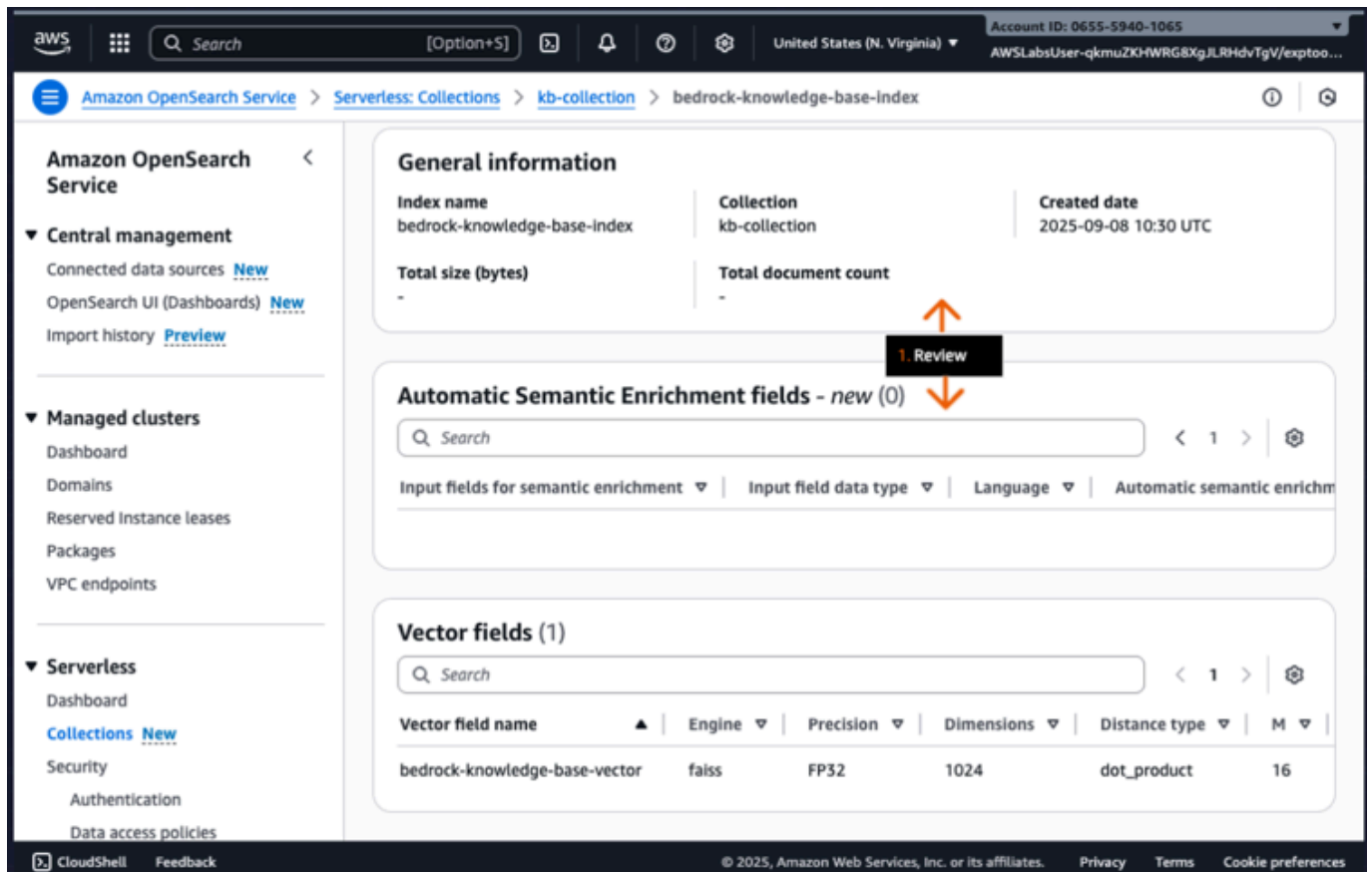
Amazon OpenSearch Service > Serverless: Collections > kb-collection

Overview | Monitor | **Indexes** | Tags

Indexes (1)

Indexing is the method by which search engines organize data for fast retrieval. [Learn more](#)

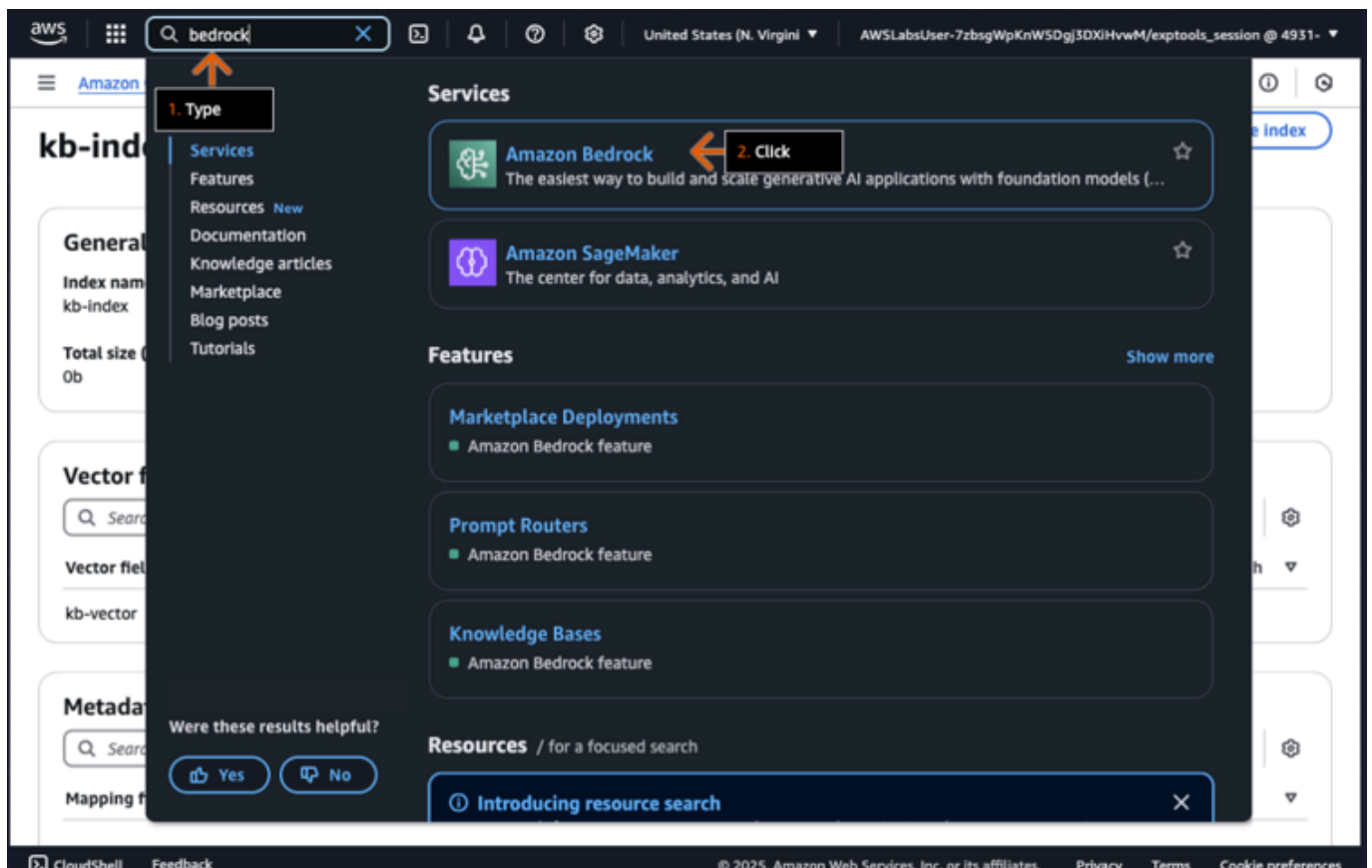
Index name	Size (bytes)	Document count	Vector field count	Auto
bedrock-knowledge-base-index	0b	0	3	0

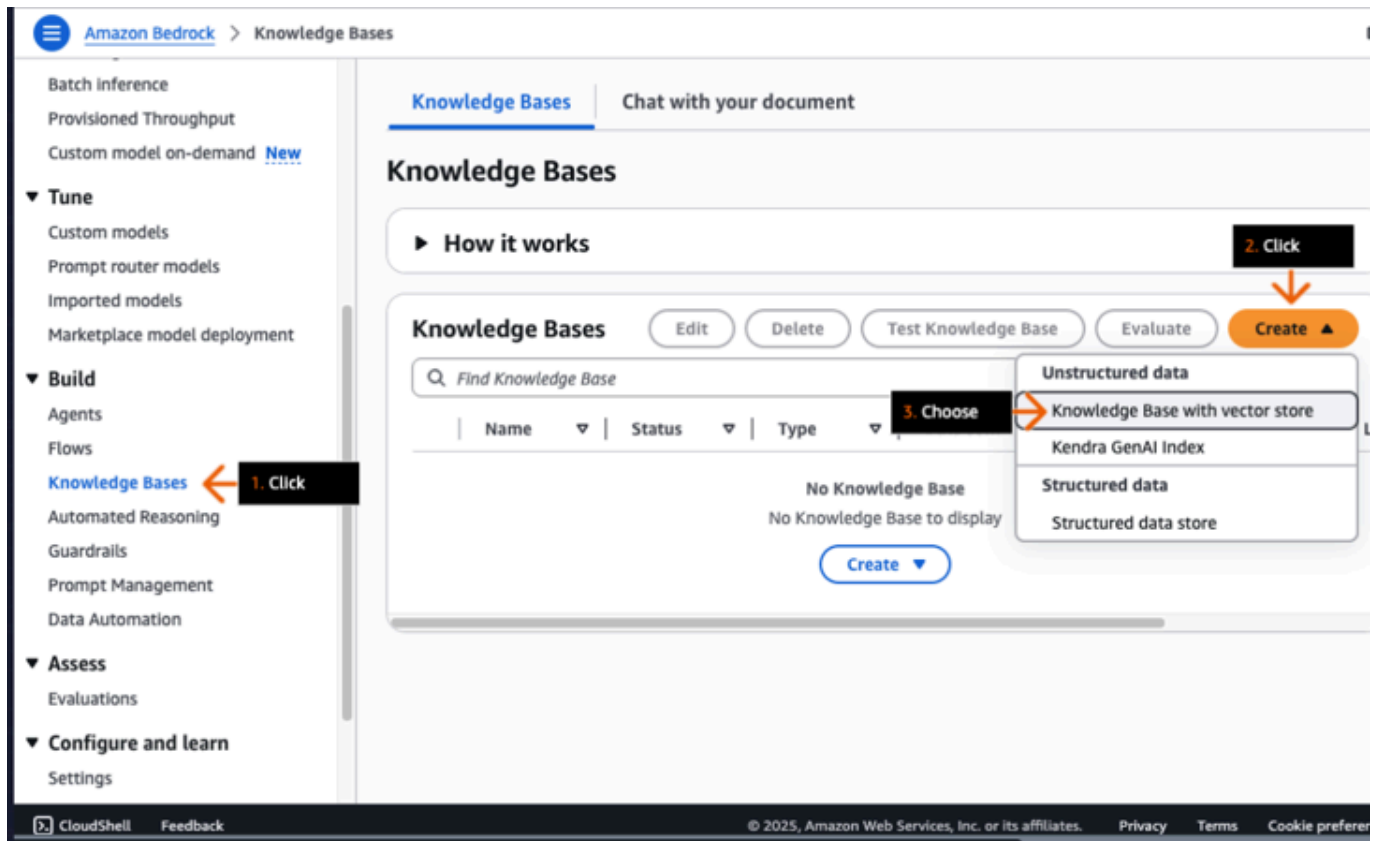


Tarea 3: Creación de la Knowledge Base (RAG)

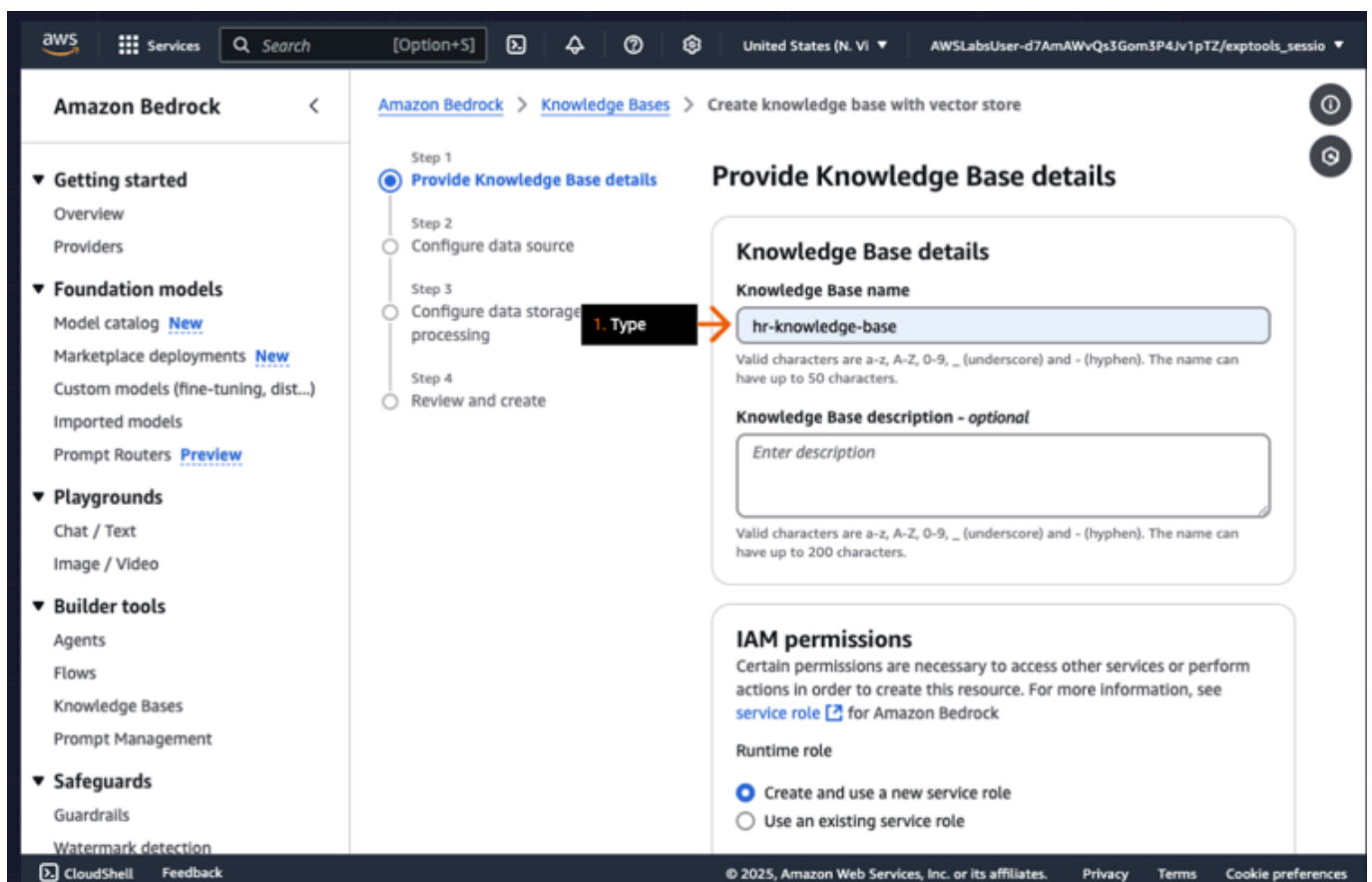
Aquí conectarás el modelo de IA con tus documentos privados.

1. **Navega a Amazon Bedrock:** Selecciona **Knowledge Bases** en el menú lateral y haz clic en **Create**.





2. **Configura los detalles:** Asigna el nombre `hr-knowledge-base` y selecciona el rol de IAM existente `Bedrock_KB_Role`.



3. **Conecta el Data Source:** Selecciona **Amazon S3** y apunta al bucket que contiene los manuales de HR.

aws Search [Option+S] United States (N. Virginia) Account ID: 2239-8179-5385 AWSLabUser-ev8NUvYouEqCHxMosNP4sp/exptool...

Amazon Bedrock > Knowledge Bases > Create knowledge base with vector store

IAM permissions

Certain permissions are necessary to access other services or perform actions in order to create this resource. For more information, see [service role](#) for Amazon Bedrock.

Runtime role

1. Choose ☐ Create and use a new service role ☒ Use an existing service role

2. Choose Choose an existing role or create a new one
To learn more, visit the section [Create a service role and configure permissions](#) in our [documentation](#).

Bedrock_KB_Role

3. Choose ☒ Amazon S3
Object storage service that stores data as objects within general purpose buckets. Add up to 5 buckets as data sources.

4. Scroll Web Crawler - Preview
Web page crawler that extracts content from public web pages you are authorized to crawl.

Custom
Create a data source directly in Amazon Bedrock. A custom data source allows the flexibility to automatically ingest documents into your vector database directly.

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aws Search [Option+S] United States (N. Virginia) Account ID: 2239-8179-5385 AWSLabUser-ev8NUvYouEqCHxMosNP4sp/exptool...

Amazon Bedrock > Knowledge Bases > Create knowledge base with vector store

Step 1 Provide Knowledge Base details

Step 2 **Configure data source**

Step 3 Configure data storage and processing

Step 4 Review and create

Configure data source

Configure for the chosen data source

Amazon S3
Provide details to connect Amazon Bedrock to your S3 data source.

Data source: hr-data-source Delete

Data source name
hr-data-source **1. Type**
Valid characters are a-z, A-Z, 0-9, _ (underscore) and - (hyphen). The name can have up to 100 characters.

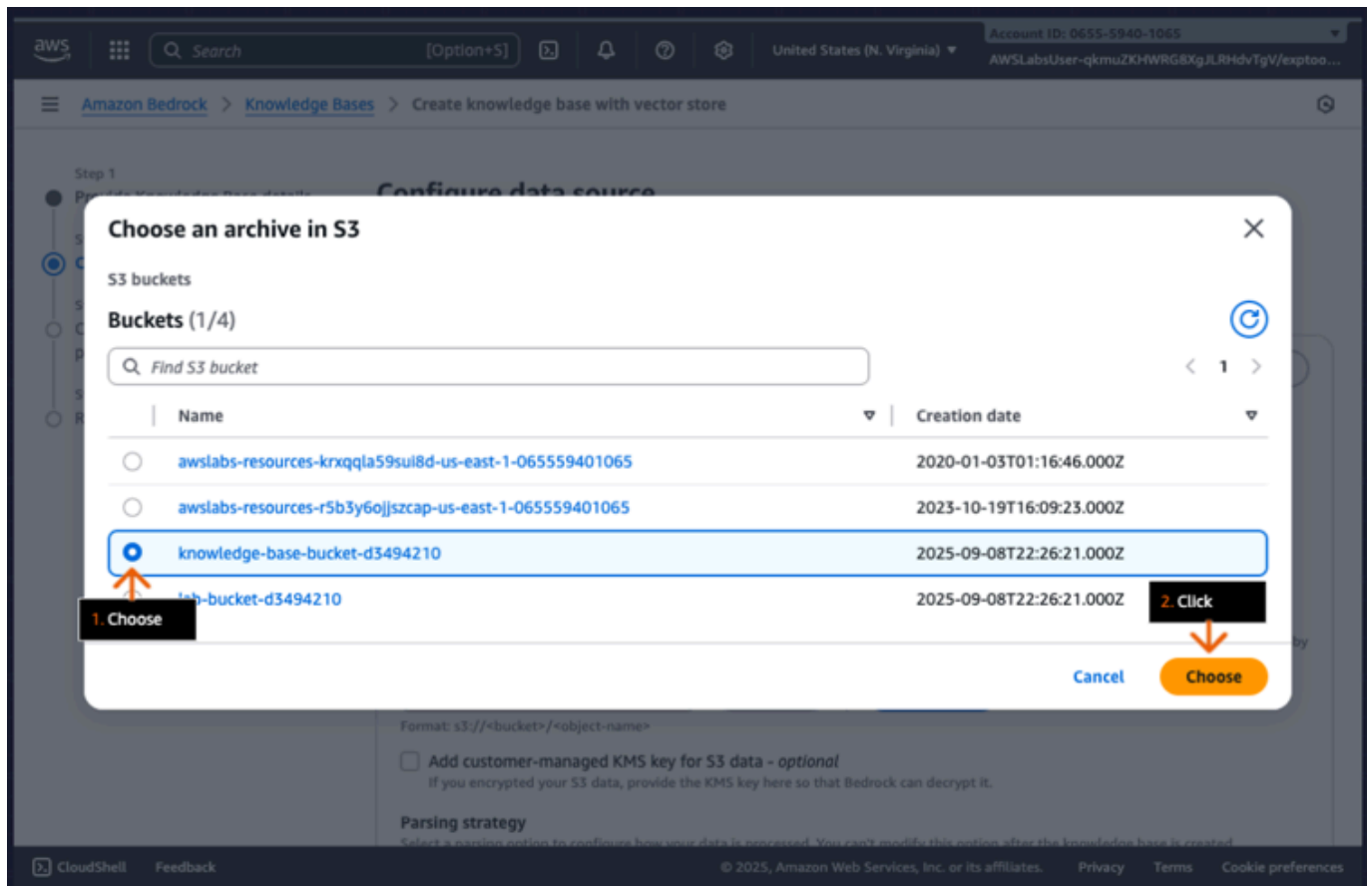
Data source location
☒ This AWS account
☐ Other AWS account

S3 URI [Info](#)
Enter the S3 bucket location containing your source files. To enhance search relevance and filtering capabilities, you can include metadata by adding a .metadata.json file for each source file.
s3://bucket-name/prefix/object **2. Click** Browse S3
Format: s3://<bucket>/<object-name>

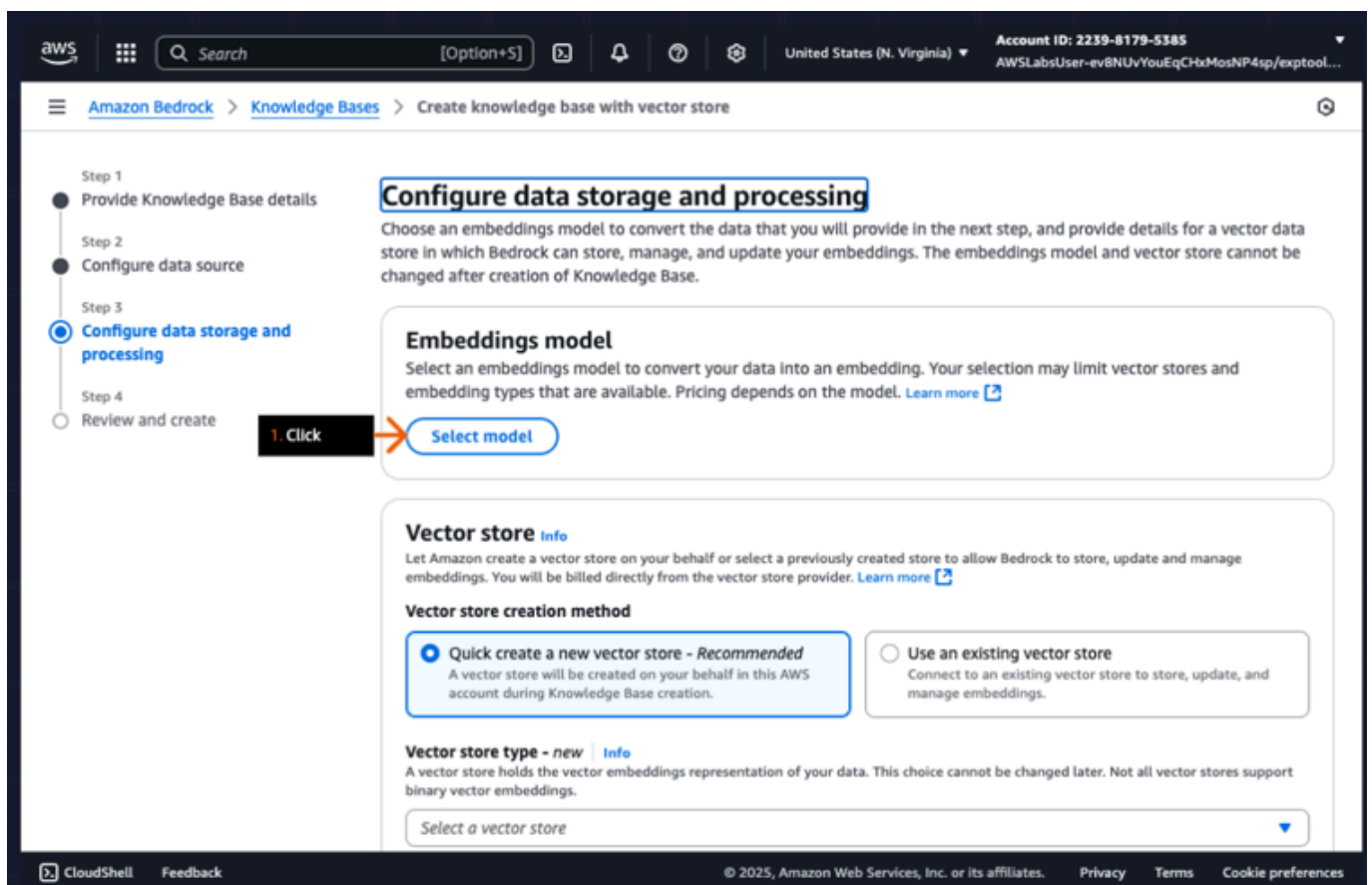
☐ Add customer-managed KMS key for S3 data - optional
If you encrypted your S3 data, provide the KMS key here so that Bedrock can decrypt it.

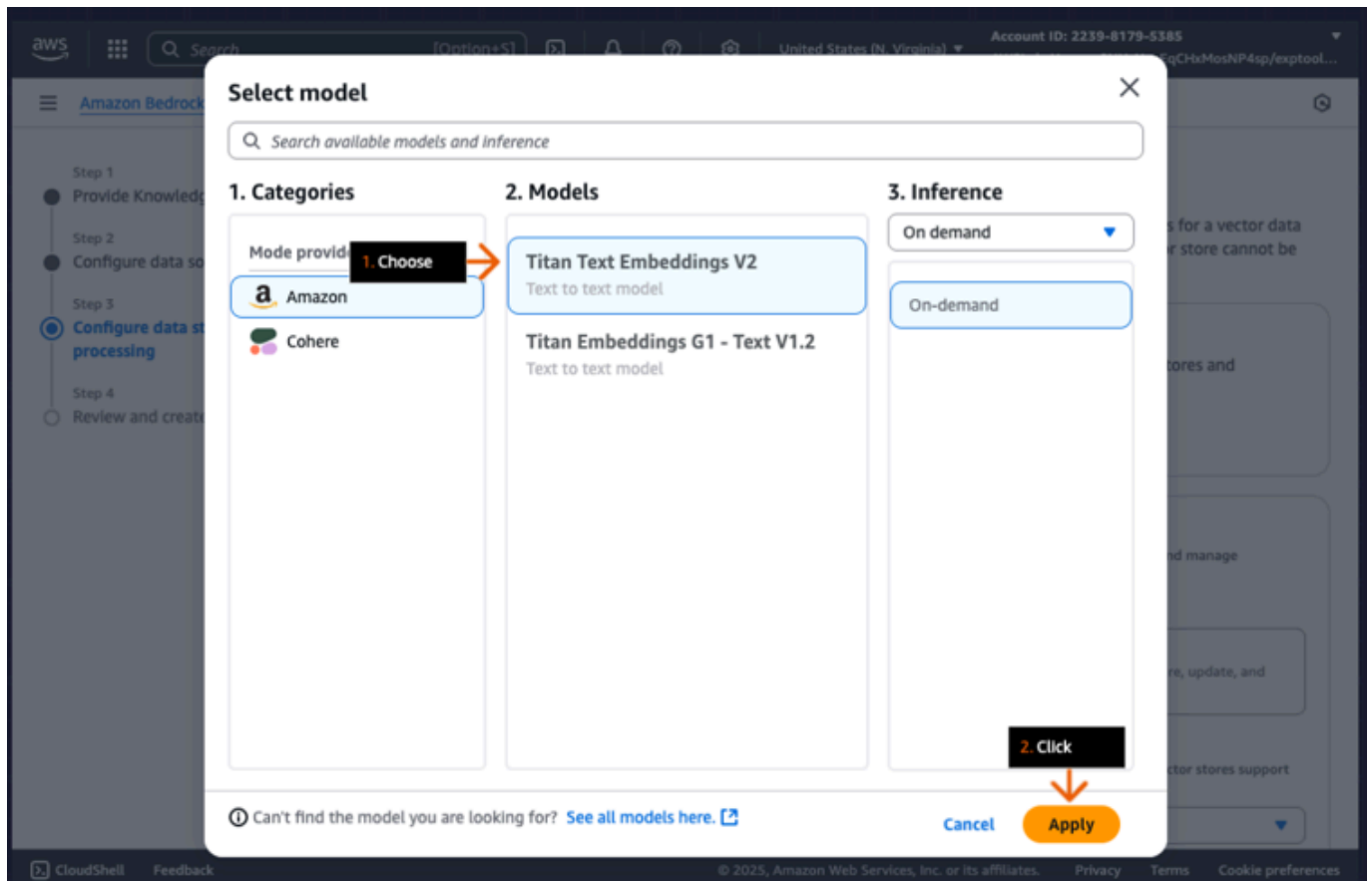
Parsing strategy
Select a parsing option to configure how your data is processed. You can't modify this option after the knowledge base is created.

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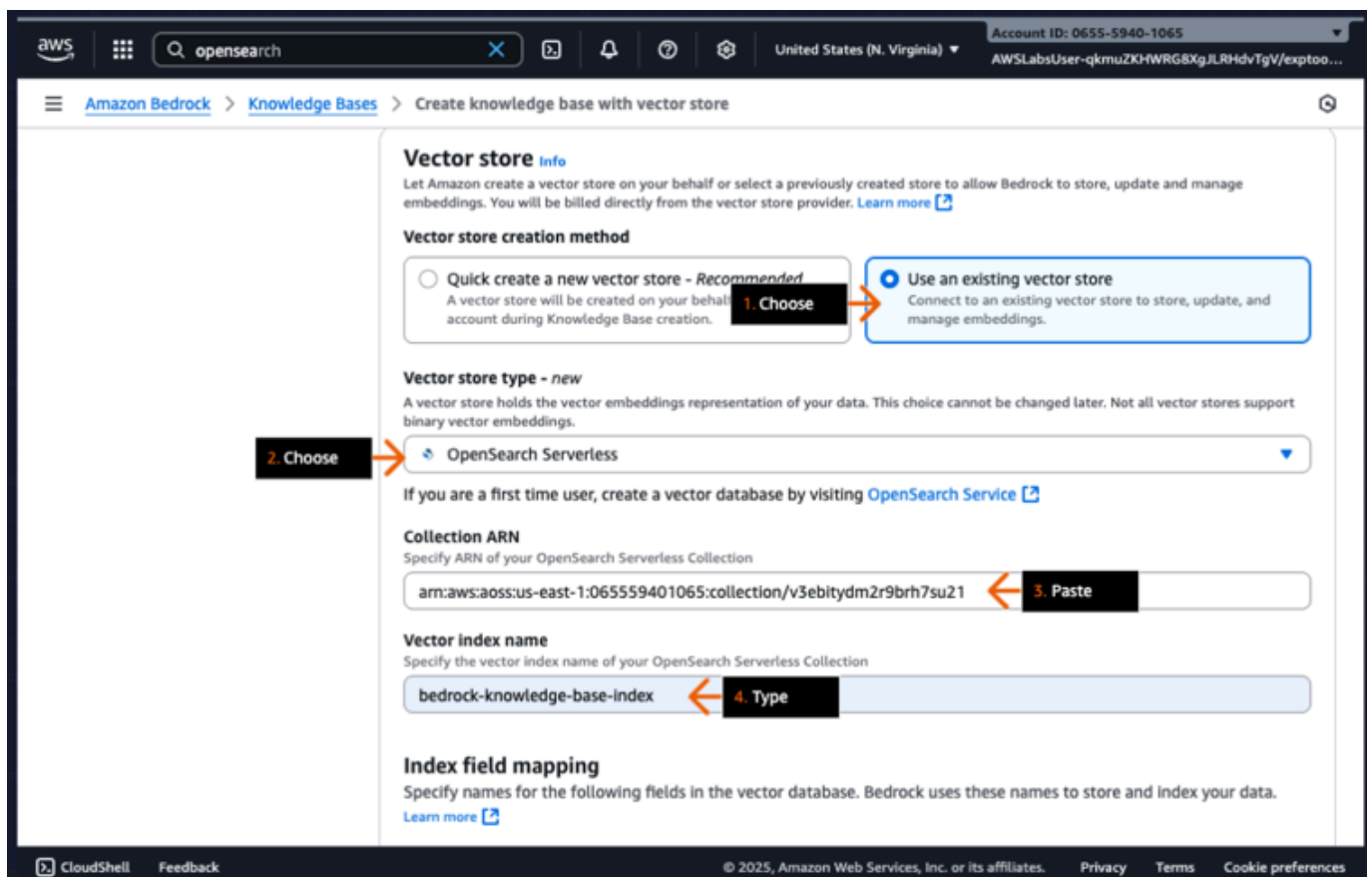


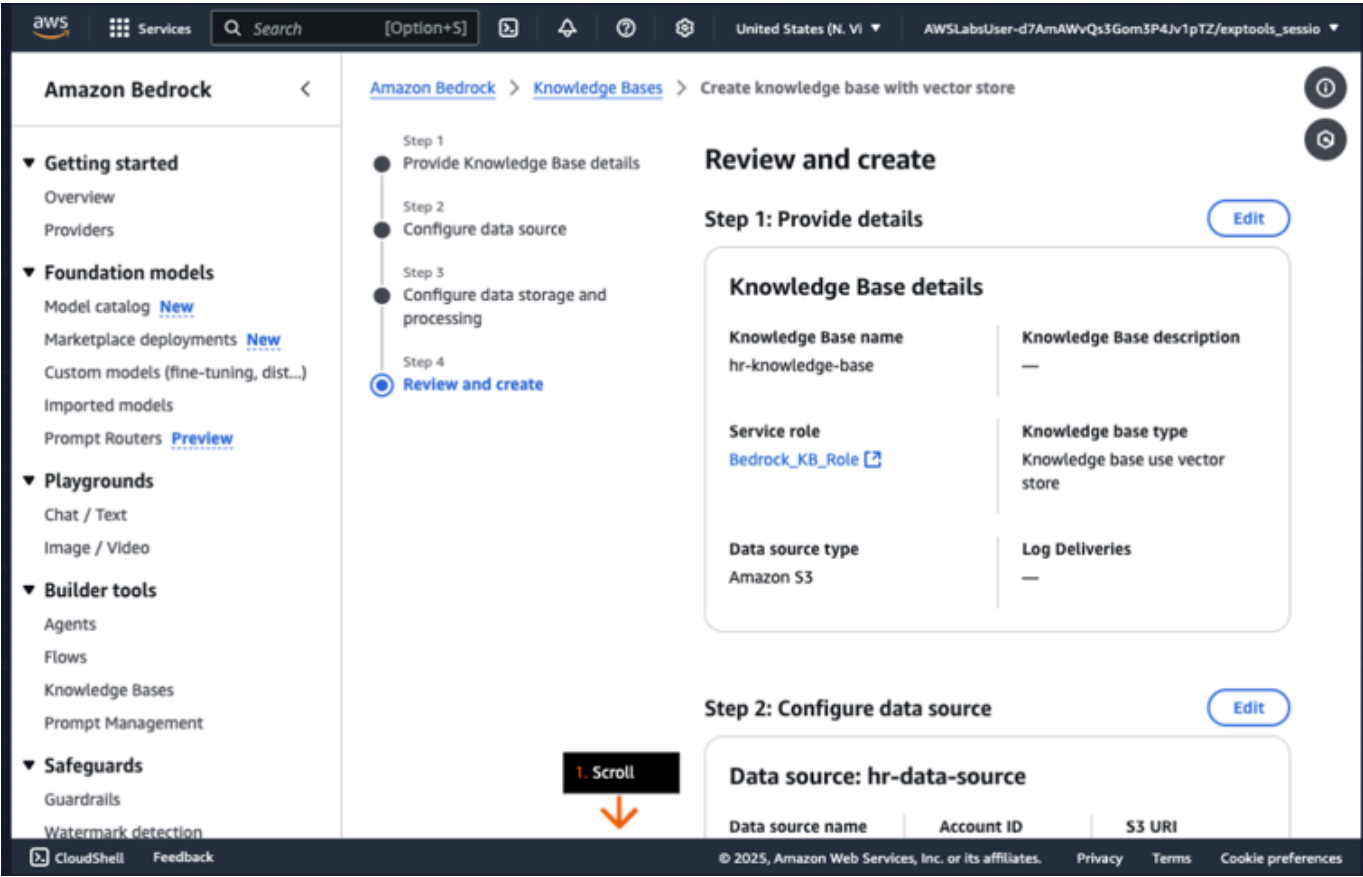
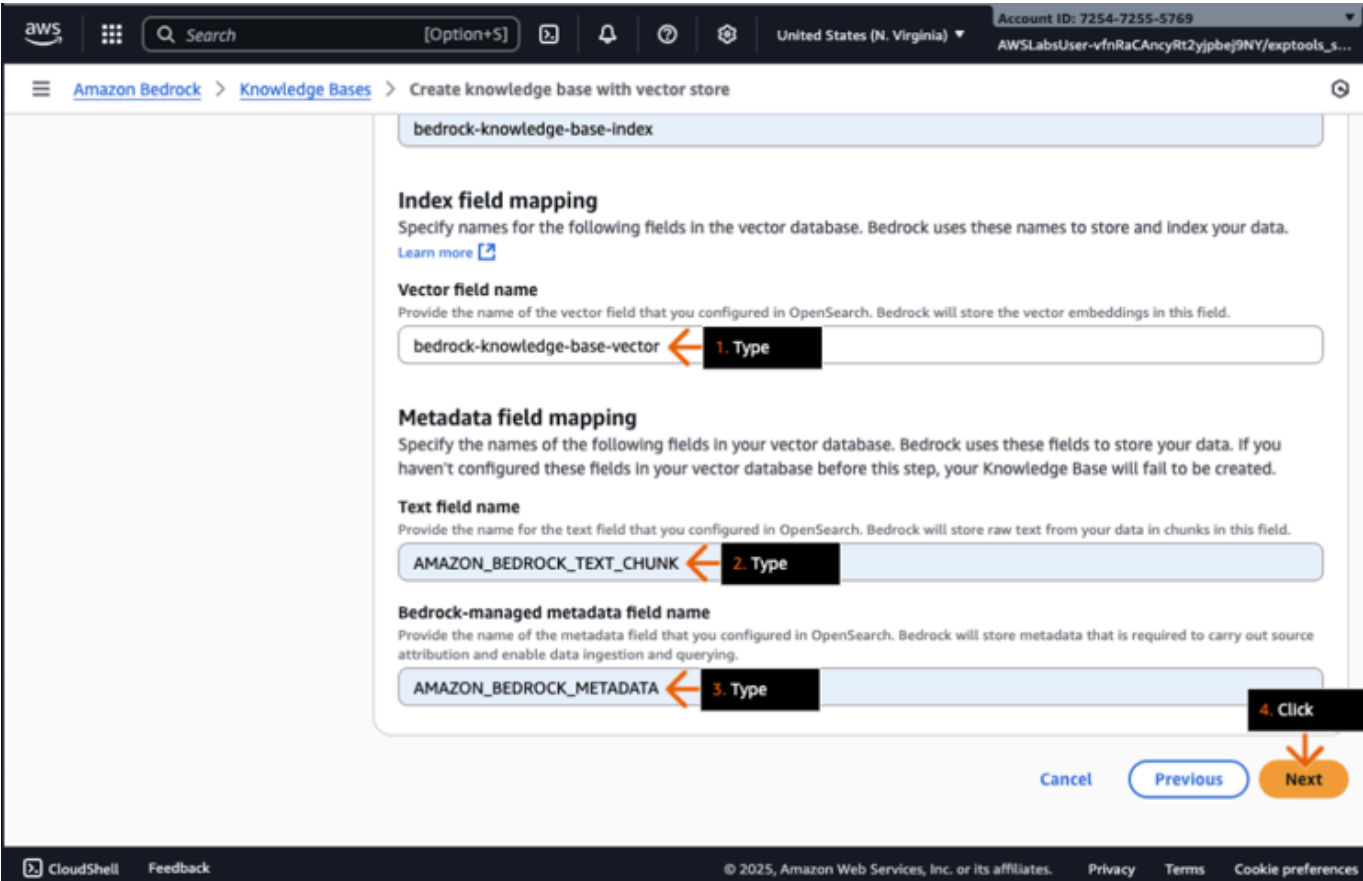
4. Selecciona el modelo de Embeddings: Elige Titan Text Embeddings V2.

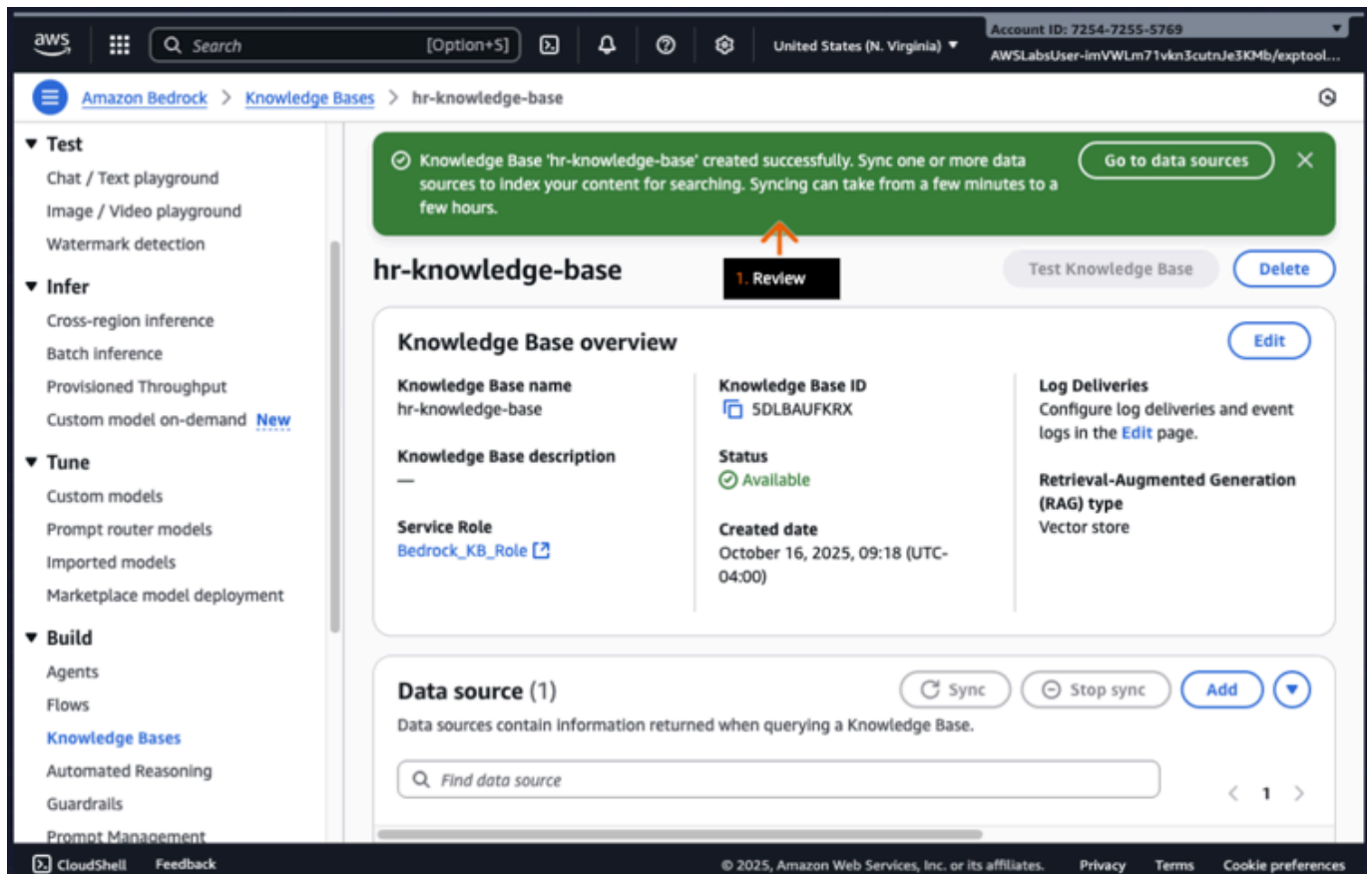




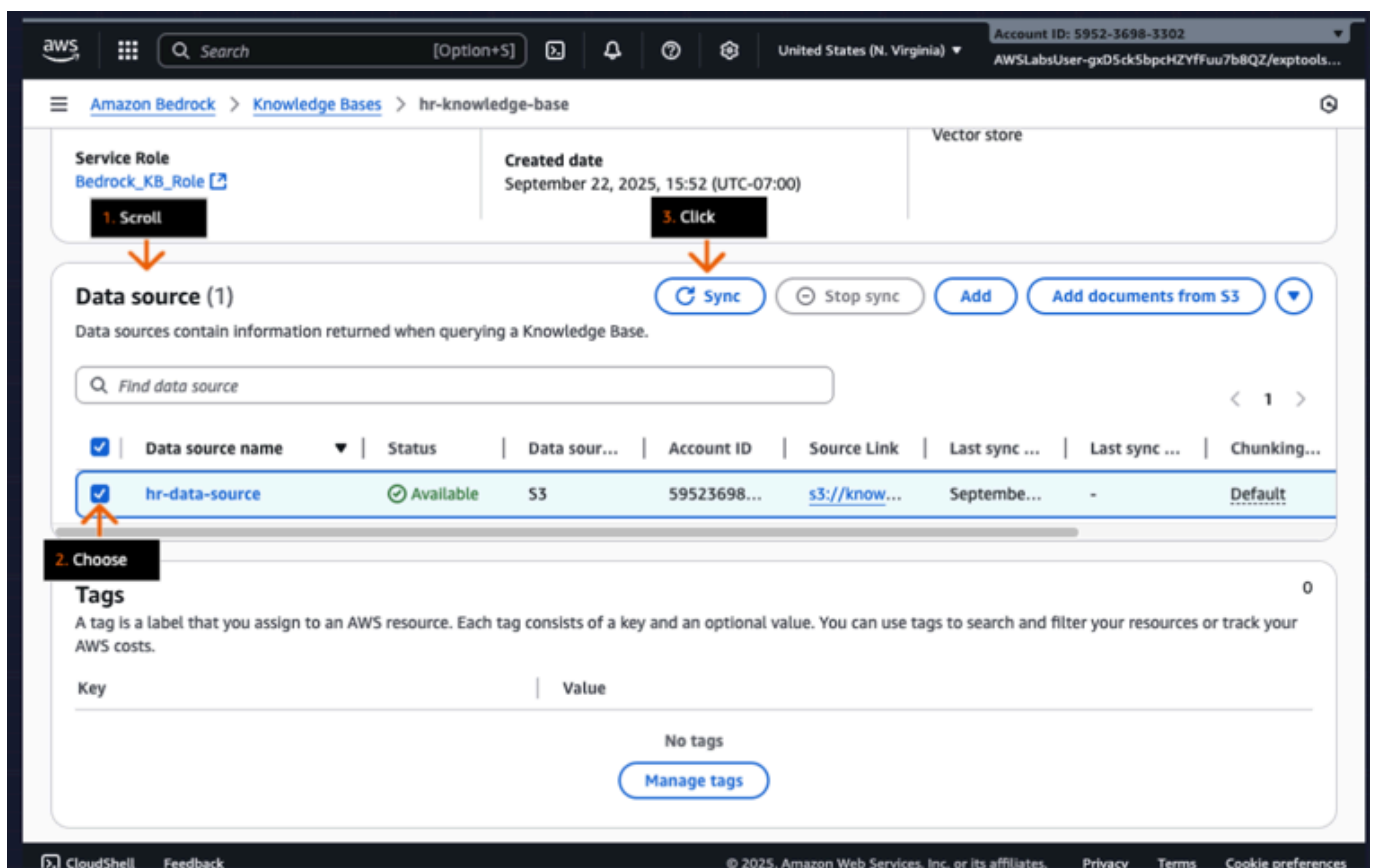
5. **Vincula el Vector Store:** Selecciona **OpenSearch Serverless**, pega el ARN de la colección y define los nombres de los campos de metadatos y vectores según las especificaciones del lab.







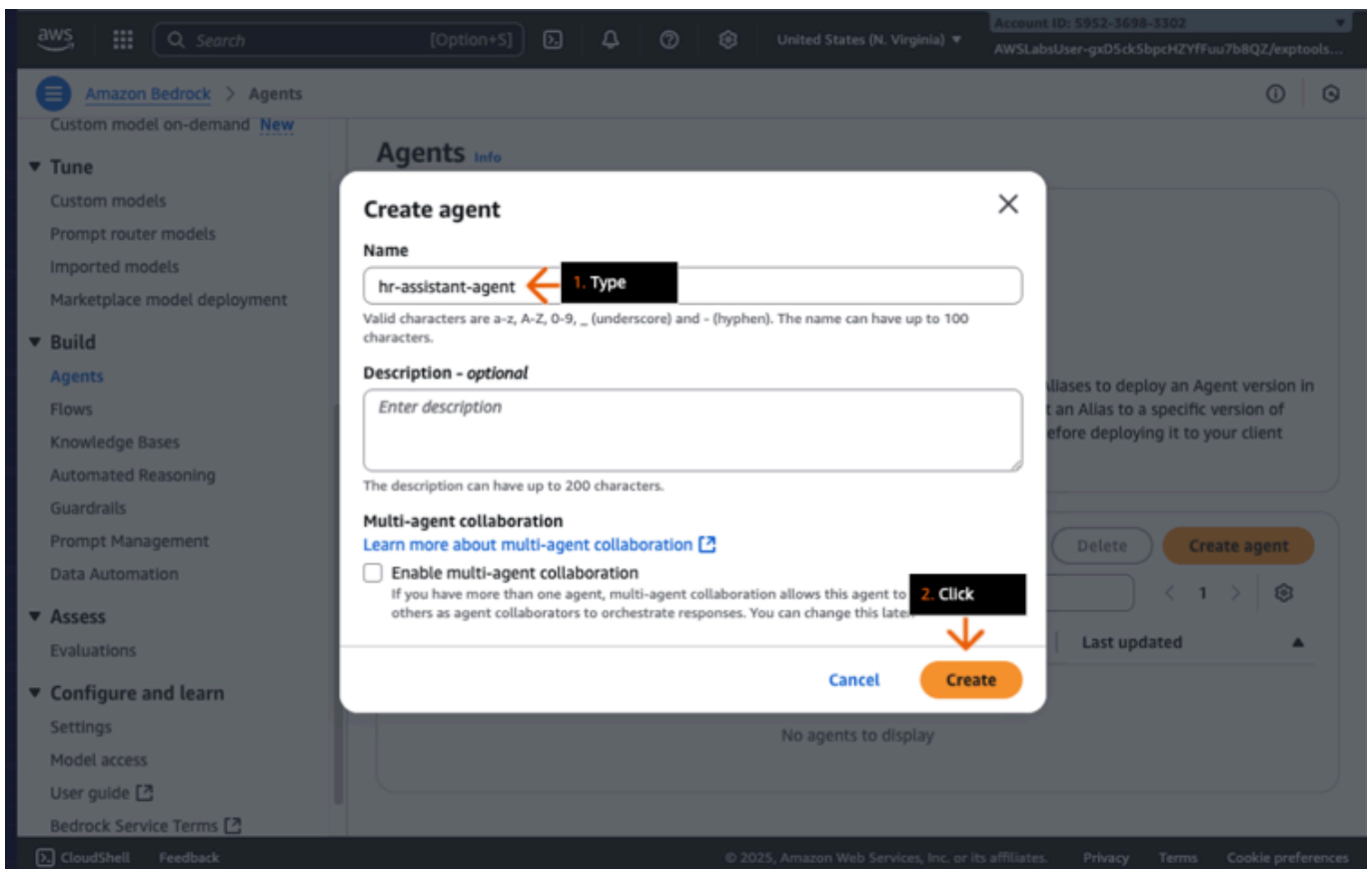
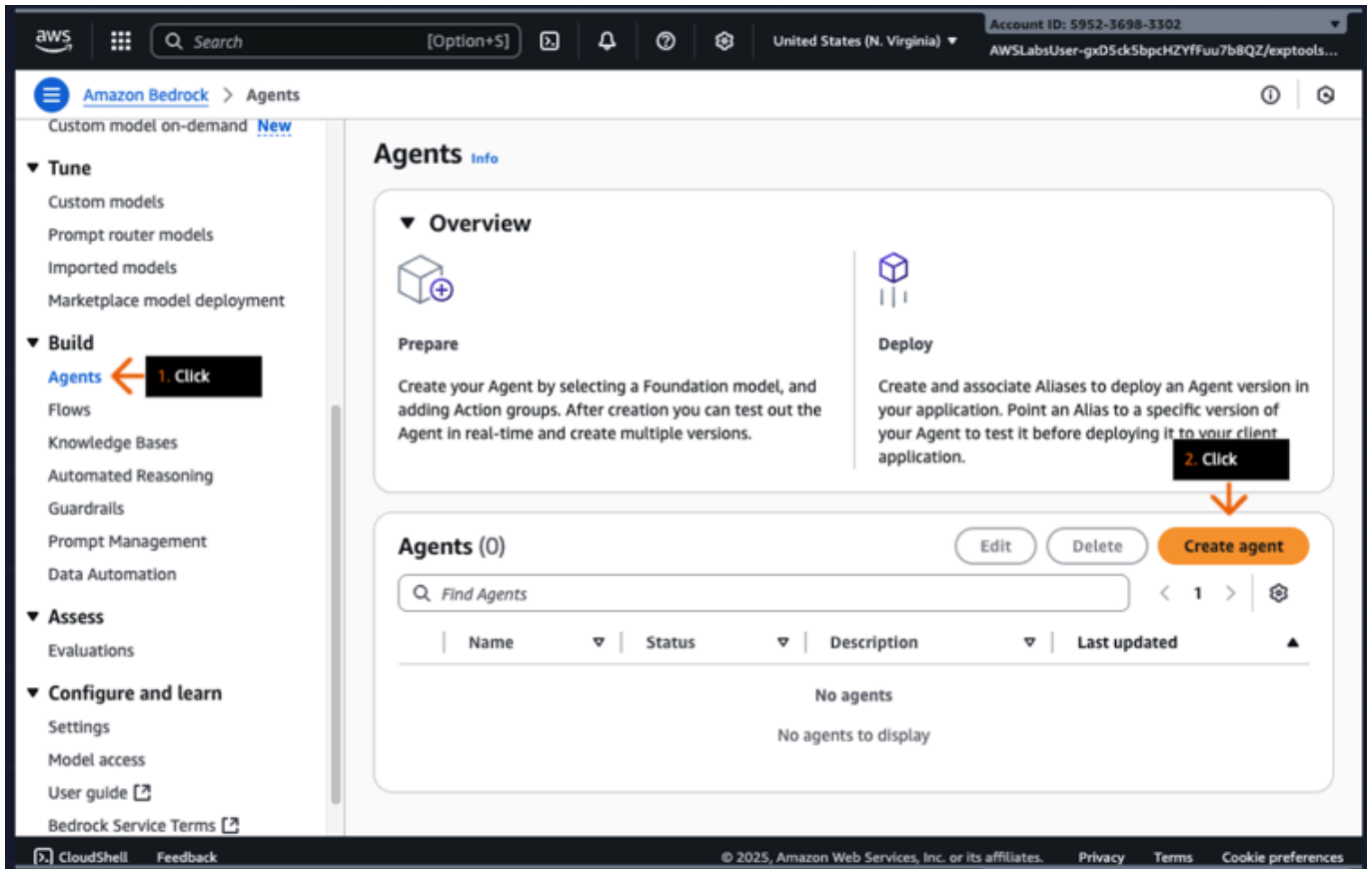
6. **Sincroniza:** Una vez creada, haz clic en **Sync** para que Bedrock procese los documentos y los convierta en vectores.

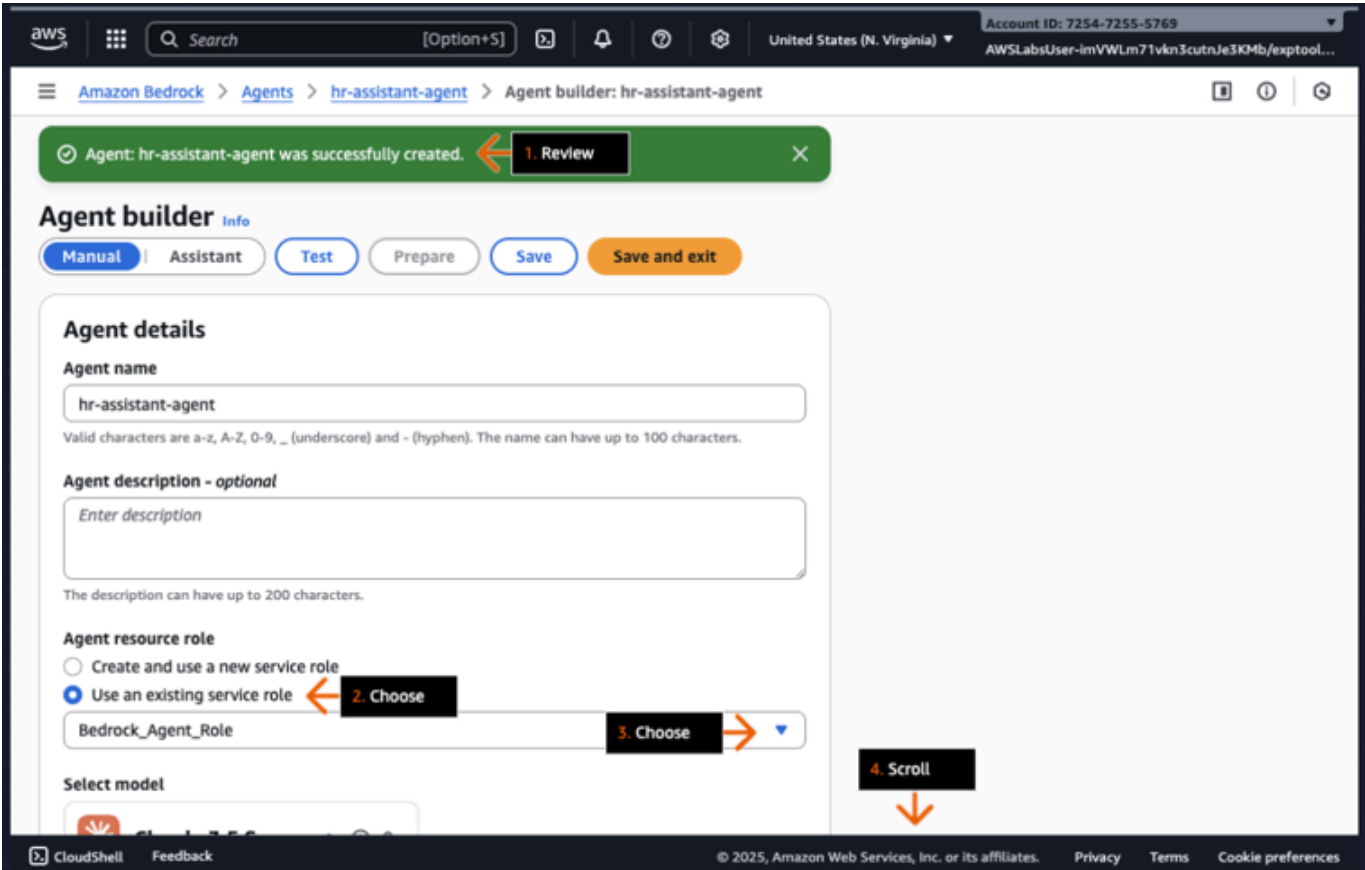


Tarea 4: Orquestación del Bedrock Agent

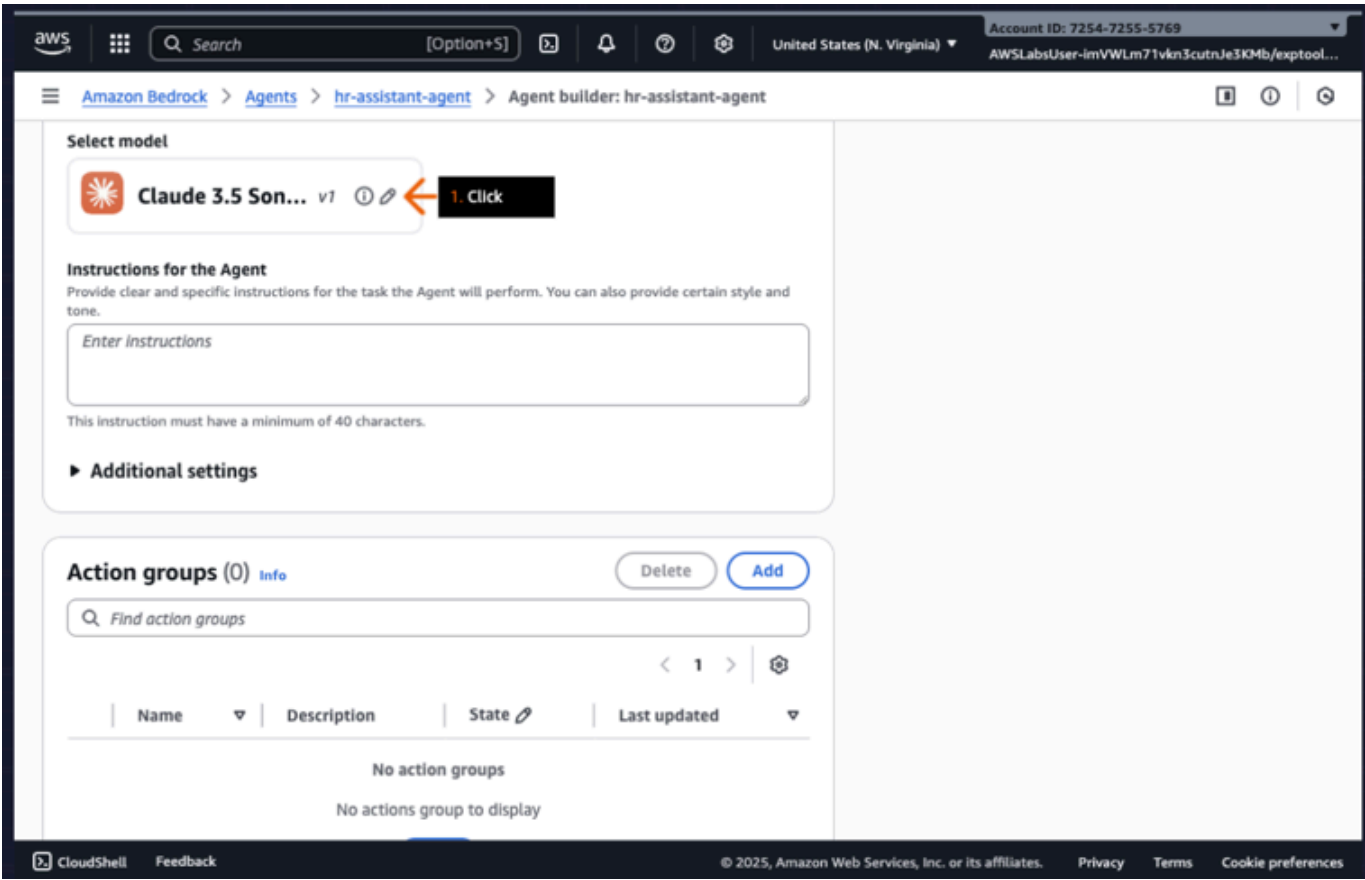
El Agente es el "cerebro" que decide si responder una pregunta o ejecutar una acción.

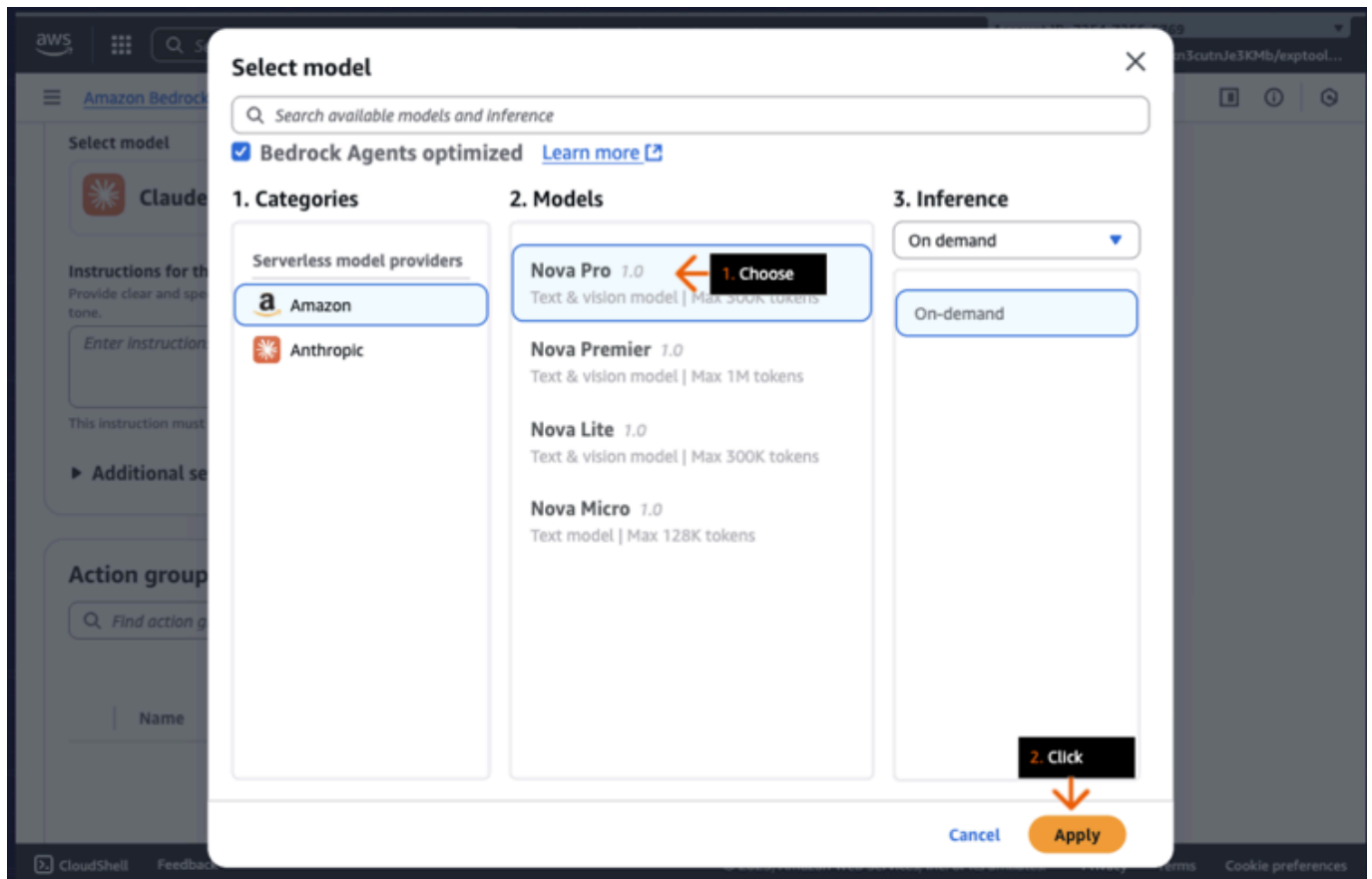
1. **Crea el Agente:** En Bedrock, ve a **Agents > Create agent**. Nombre: **hr-assistant-agent**.



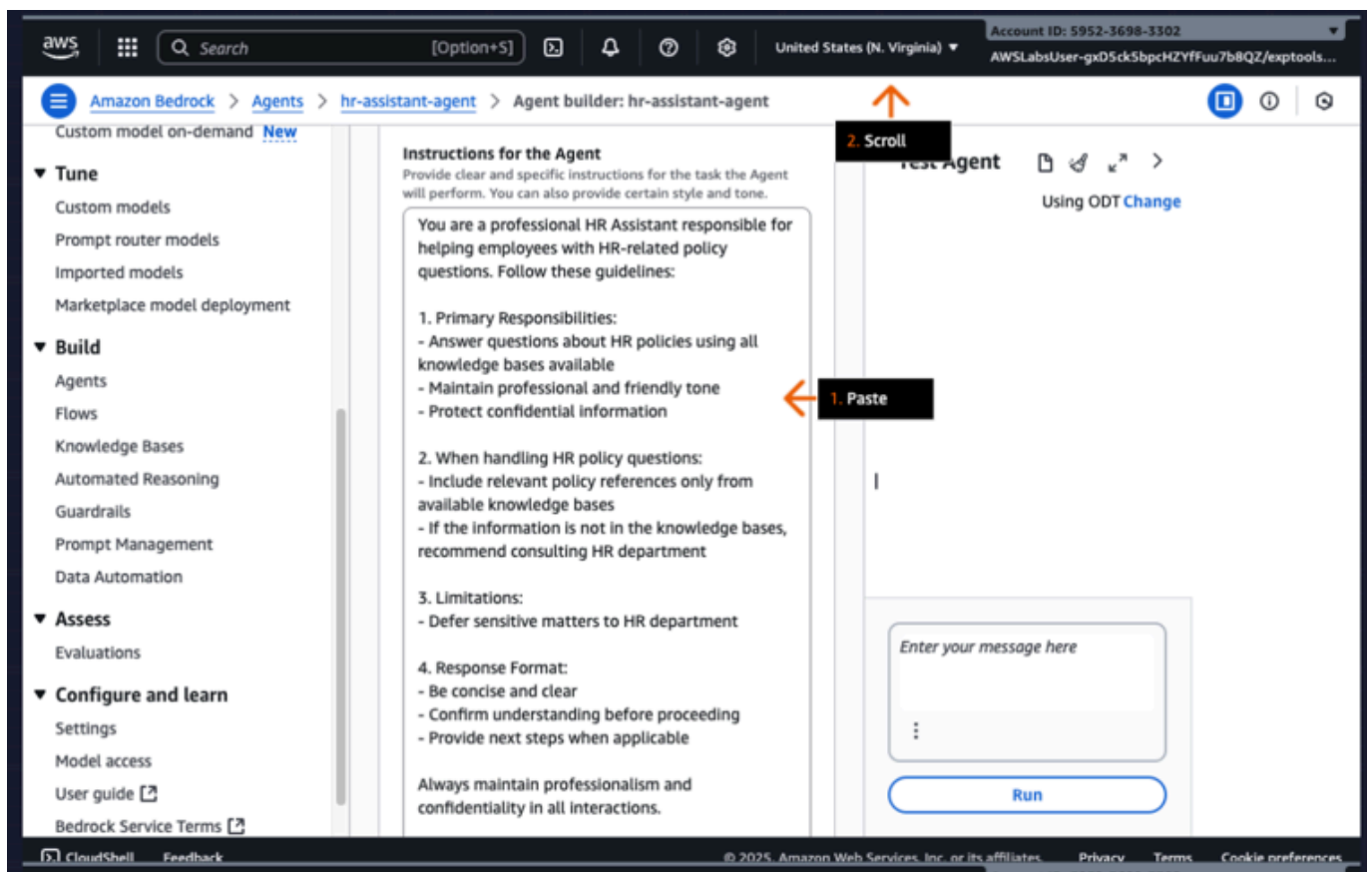


2. **Selecciona el Modelo:** Elige **Amazon Nova Pro 1.0** (optimizado para agentes inteligentes).

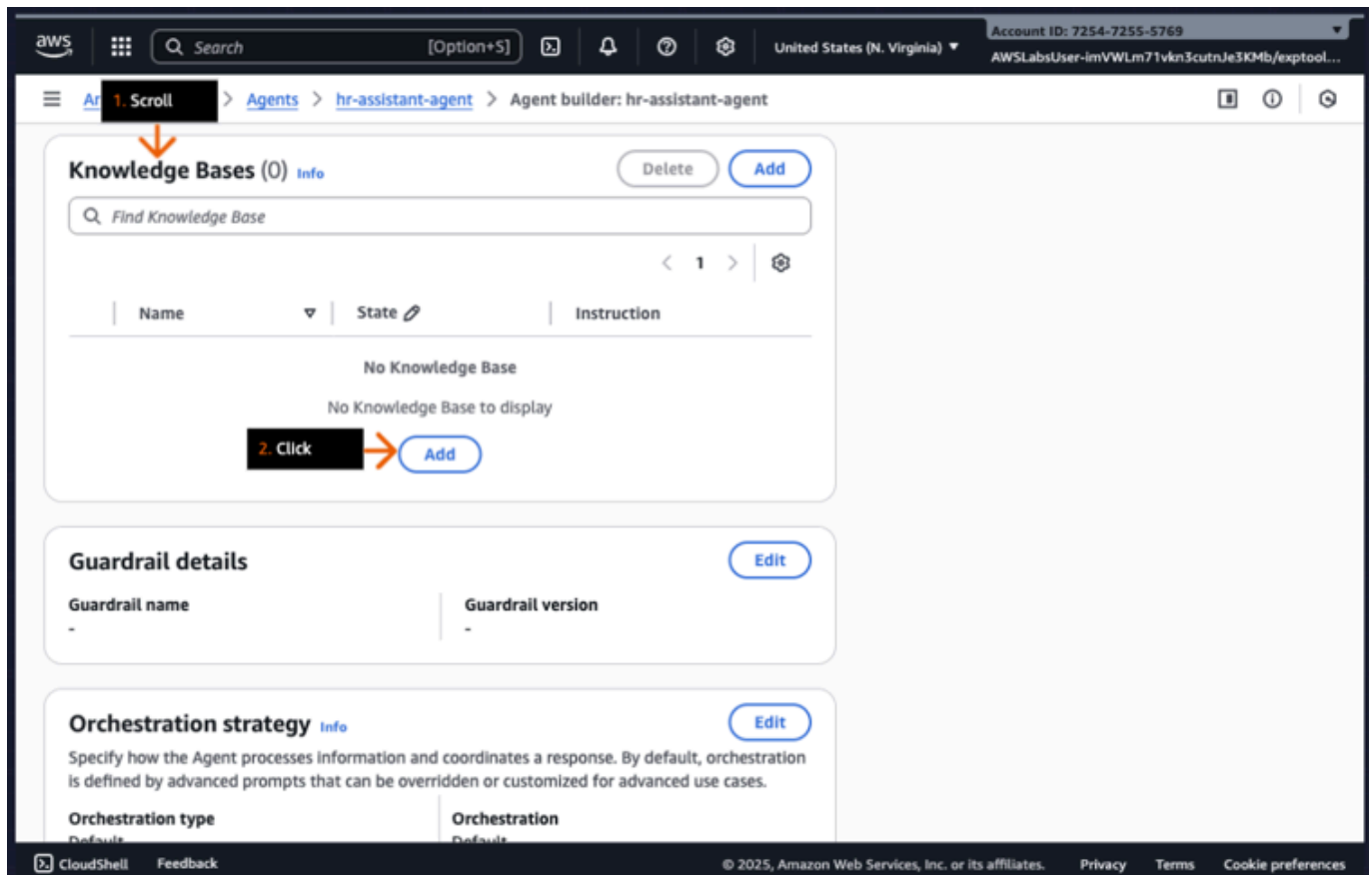




3. **Configura las Instrucciones:** Pega el contenido que copiaste del archivo `agent-prompt.txt`.



4. **Asocia la Knowledge Base:** En la sección de Knowledge Bases del agente, añade `hr-knowledge-base`. Esto le da acceso a los manuales de la empresa.



Knowledge Bases (0) [Info](#) [Delete](#) [Add](#)

< 1 > [Settings](#)

Name	State	Instruction
No Knowledge Base		
No Knowledge Base to display		

2. Click [Add](#)

Guardrail details [Edit](#)

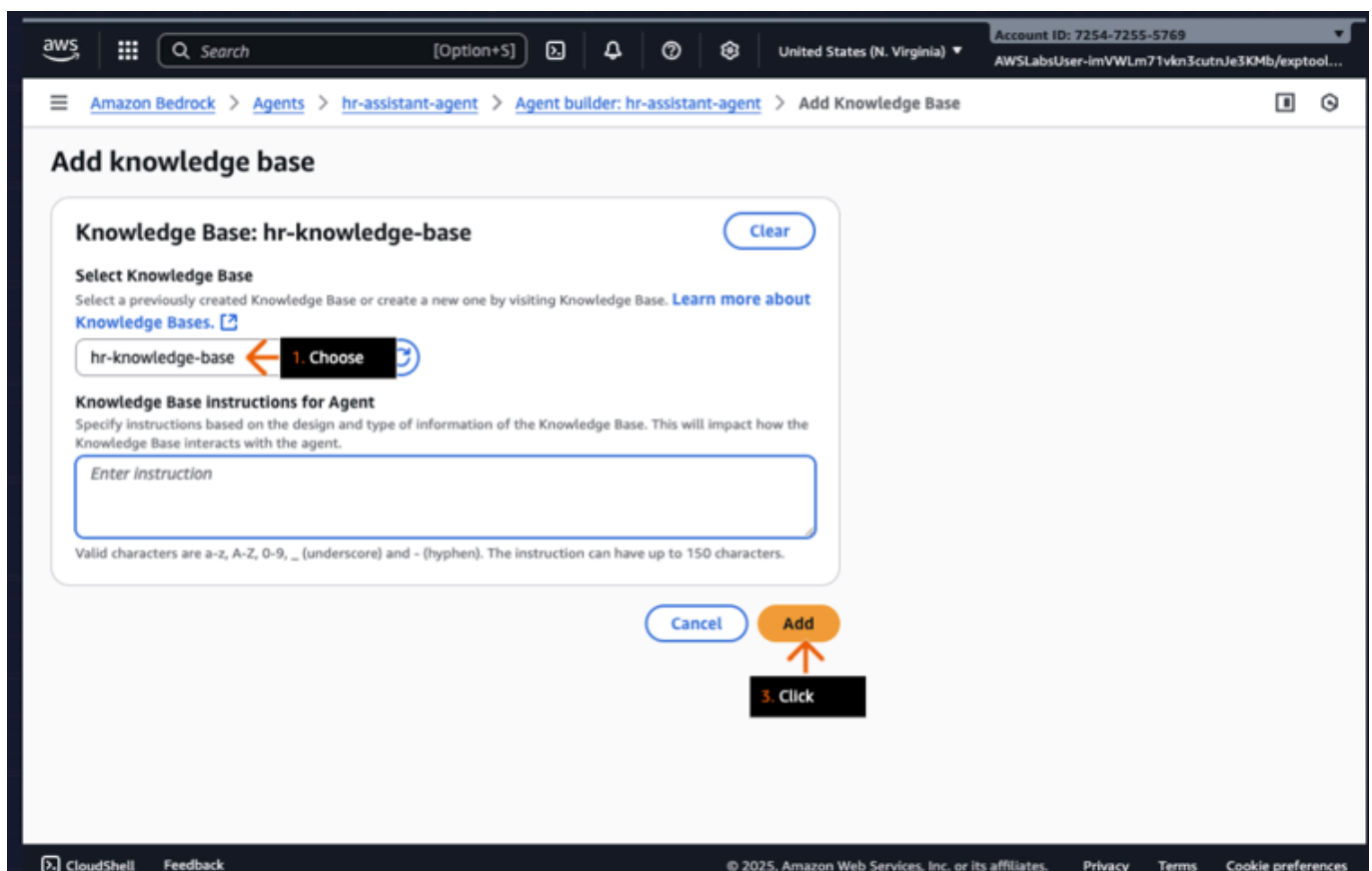
Guardrail name: - Guardrail version: -

Orchestration strategy [Info](#) [Edit](#)

Specify how the Agent processes information and coordinates a response. By default, orchestration is defined by advanced prompts that can be overridden or customized for advanced use cases.

Orchestration type: Default Orchestration: Default

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Add knowledge base

Knowledge Base: hr-knowledge-base [Clear](#)

Select Knowledge Base

Select a previously created Knowledge Base or create a new one by visiting Knowledge Base. [Learn more about Knowledge Bases.](#)

hr-knowledge-base **1. Choose**

Knowledge Base instructions for Agent

Specify instructions based on the design and type of information of the Knowledge Base. This will impact how the Knowledge Base interacts with the agent.

Valid characters are a-z, A-Z, 0-9, _ (underscore) and - (hyphen). The instruction can have up to 150 characters.

[Cancel](#) [Add](#)

3. Click

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Amazon Bedrock > Agents > hr-assistant-agent > Agent builder: hr-assistant-agent

Custom model on-demand [New](#)

▼ Tune

- Custom models
- Prompt router models
- Imported models
- Marketplace model deployment

▼ Build

- Agents
- Flows
- Knowledge Bases
- Automated Reasoning
- Guardrails
- Prompt Management
- Data Automation

▼ Assess

- Evaluations

▼ Configure and learn

- Settings
- Model access
- User guide [📖](#)
- Bedrock Service Terms [📄](#)

Agent builder Info

[Manual](#) | [Assistant](#) | [Test](#) | [Prepare](#)

[Save](#) [Save and exit](#)

Agent details

Agent name

hr-assistant-agent

Valid characters are a-z, A-Z, 0-9, _ (underscore) and - (hyphen). The name can have up to 100 characters.

Agent description - optional

Enter description

The description can have up to 200 characters.

Agent resource role

☐ Create and use a new service role

☒ Use an existing service role

Test Agent

Using ODT [Change](#)

1. Review

2. Click

3. Click

Prepare the Agent to test the latest changes.

Prepare

Enter your message here

Run

Amazon Bedrock > Agents > hr-assistant-agent > Agent builder: hr-assistant-agent

Custom model on-demand [New](#)

▼ Tune

- Custom models
- Prompt router models
- Imported models
- Marketplace model deployment

▼ Build

- Agents
- Flows
- Knowledge Bases
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- Guardrails
- Prompt Management
- Data Automation

▼ Assess

- Evaluations

▼ Configure and learn

- Settings
- Model access
- User guide [📖](#)
- Bedrock Service Terms [📄](#)

Agent builder Info

[Manual](#) | [Assistant](#) | [Test](#) | [Prepare](#)

[Save](#) [Save and exit](#)

Agent details

Agent name

hr-assistant-agent

Valid characters are a-z, A-Z, 0-9, _ (underscore) and - (hyphen). The name can have up to 100 characters.

Agent description - optional

Enter description

The description can have up to 200 characters.

Agent resource role

☐ Create and use a new service role

☒ Use an existing service role

arn:aws:iam::595236983302:role/BedrockAgent...

Test Agent

Using ODT [Change](#)

1. Review

2. Type

3. Click

Agent: hr-assistant-agent was successfully prepared.

What are the standard work hours?

Run

Agent: hr-assistant-agent was successfully prepared.

Agent builder

Manual | Assistant | **Test** | Prepare | Save | Save and exit

Agent details

Agent name
hr-assistant-agent
Valid characters are a-z, A-Z, 0-9, _ (underscore) and - (hyphen). The name can have up to 100 characters.

Agent description - optional
Enter description
The description can have up to 200 characters.

Agent resource role
☐ Create and use a new service role
☒ Use an existing service role
arn:aws:iam::595236983302:role/Bedrock_Agent_Role

Select model

Test Agent

Using ODT [Change](#)

What are the standard work hours?

The standard work hours are 9 AM to 5 PM, Monday through Friday. All employees must be available during core hours, which are 10 AM to 3 PM. [Show trace >](#)

What is the vacation policy?

[Run](#)

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Agent: hr-assistant-agent was successfully prepared.

Agent builder

Manual | Assistant | **Test** | Prepare | Save | Save and exit

Agent details

Agent name
hr-assistant-agent
Valid characters are a-z, A-Z, 0-9, _ (underscore) and - (hyphen). The name can have up to 100 characters.

Agent description - optional
Enter description
The description can have up to 200 characters.

Agent resource role
☐ Create and use a new service role
☒ Use an existing service role
arn:aws:iam::595236983302:role/Bedrock_Agent_Role

Select model

Test Agent

Using ODT [Change](#)

What is the vacation policy?

The vacation policy is as follows: Vacation time accrues annually based on years of service. Employees with 0-2 years of service accrue 15 days, those with 3-5 years accrue 18 days, and those with 6+ years accrue 20 days. A maximum of 5 days can be rolled over per year. Vacation requests must be made at least 2 weeks in advance and are subject to manager approval. There are

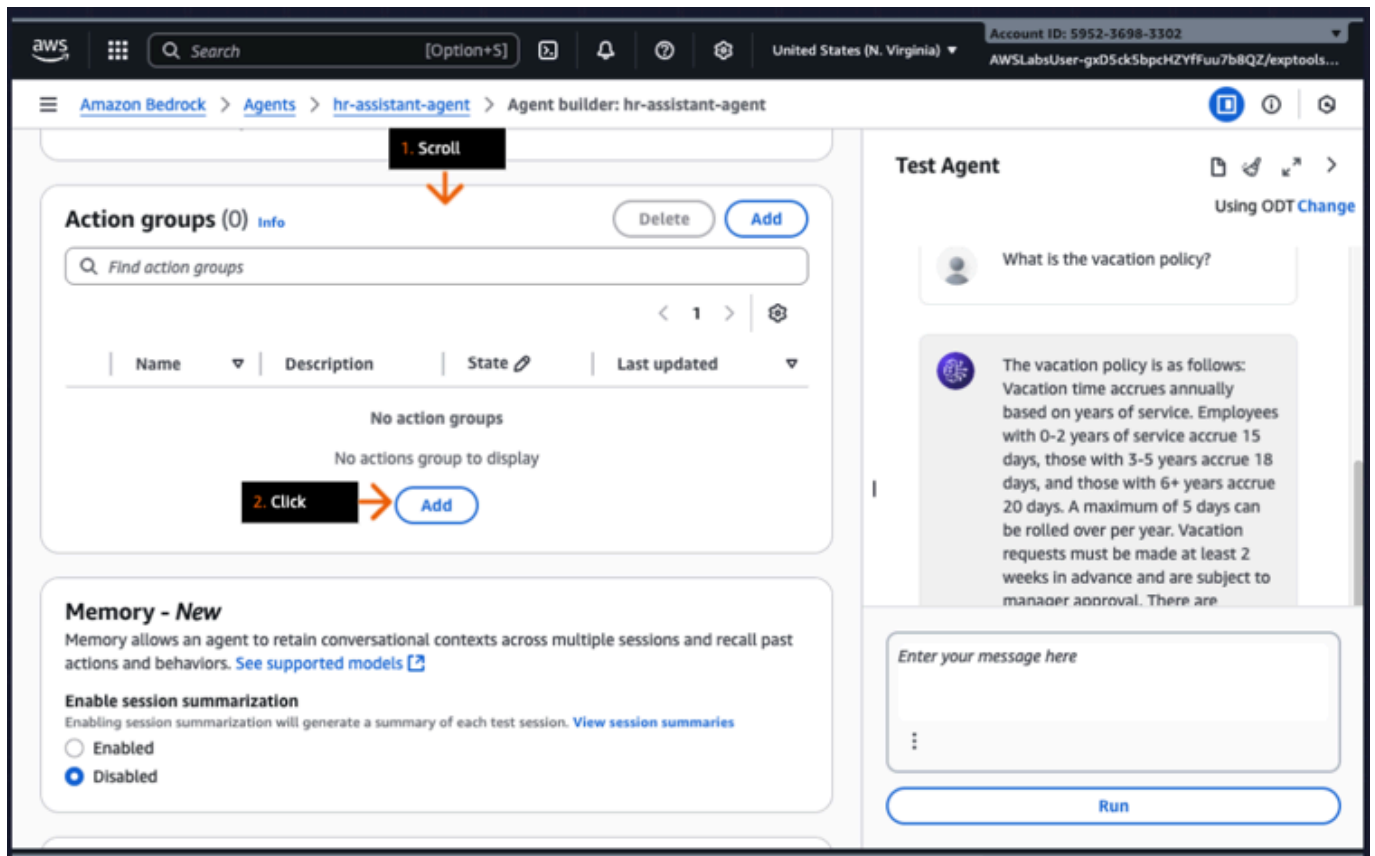
Enter your message here

[Run](#)

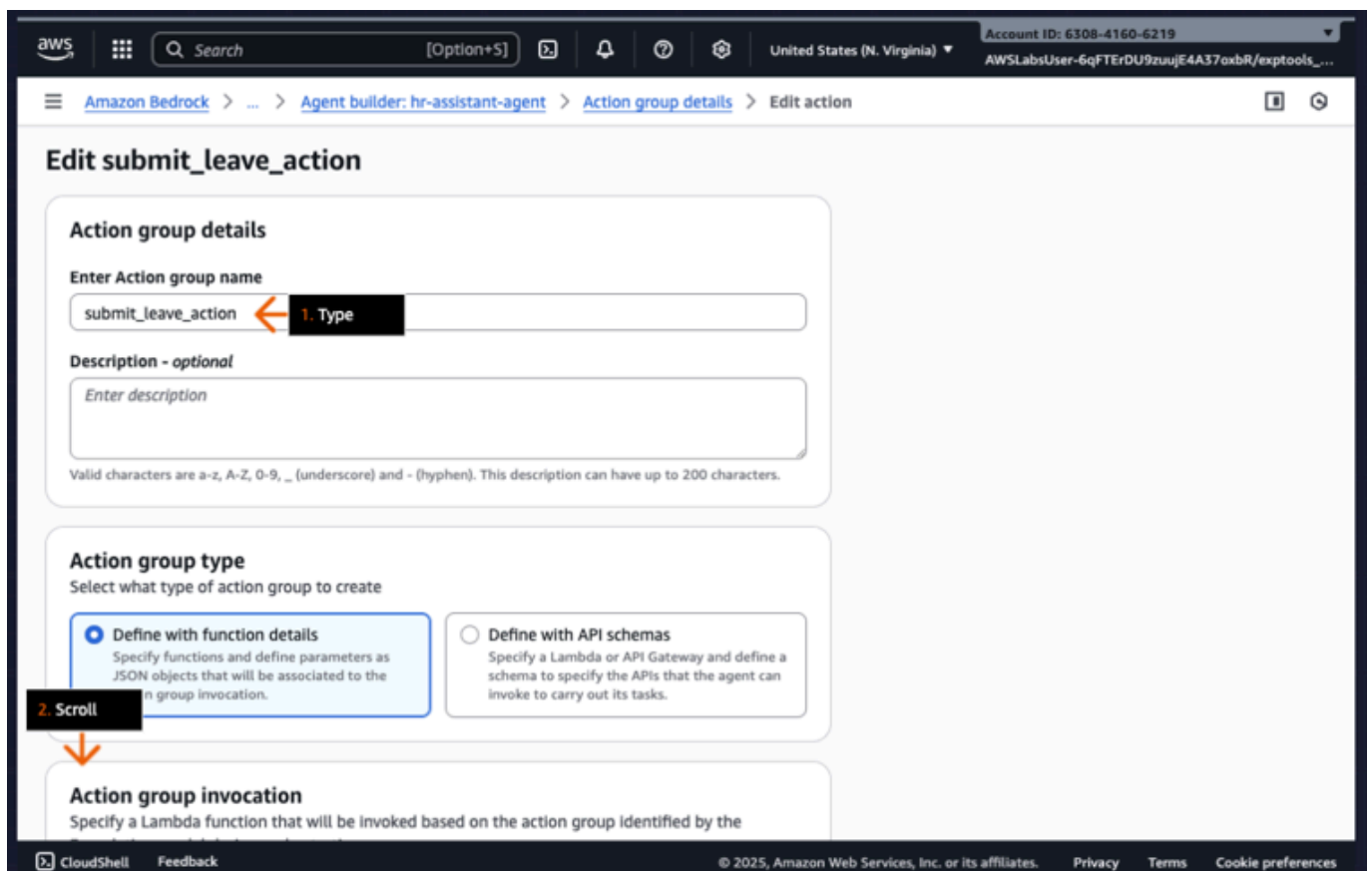
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Tarea 5: Implementación de Action Groups (Lógica de Negocio)

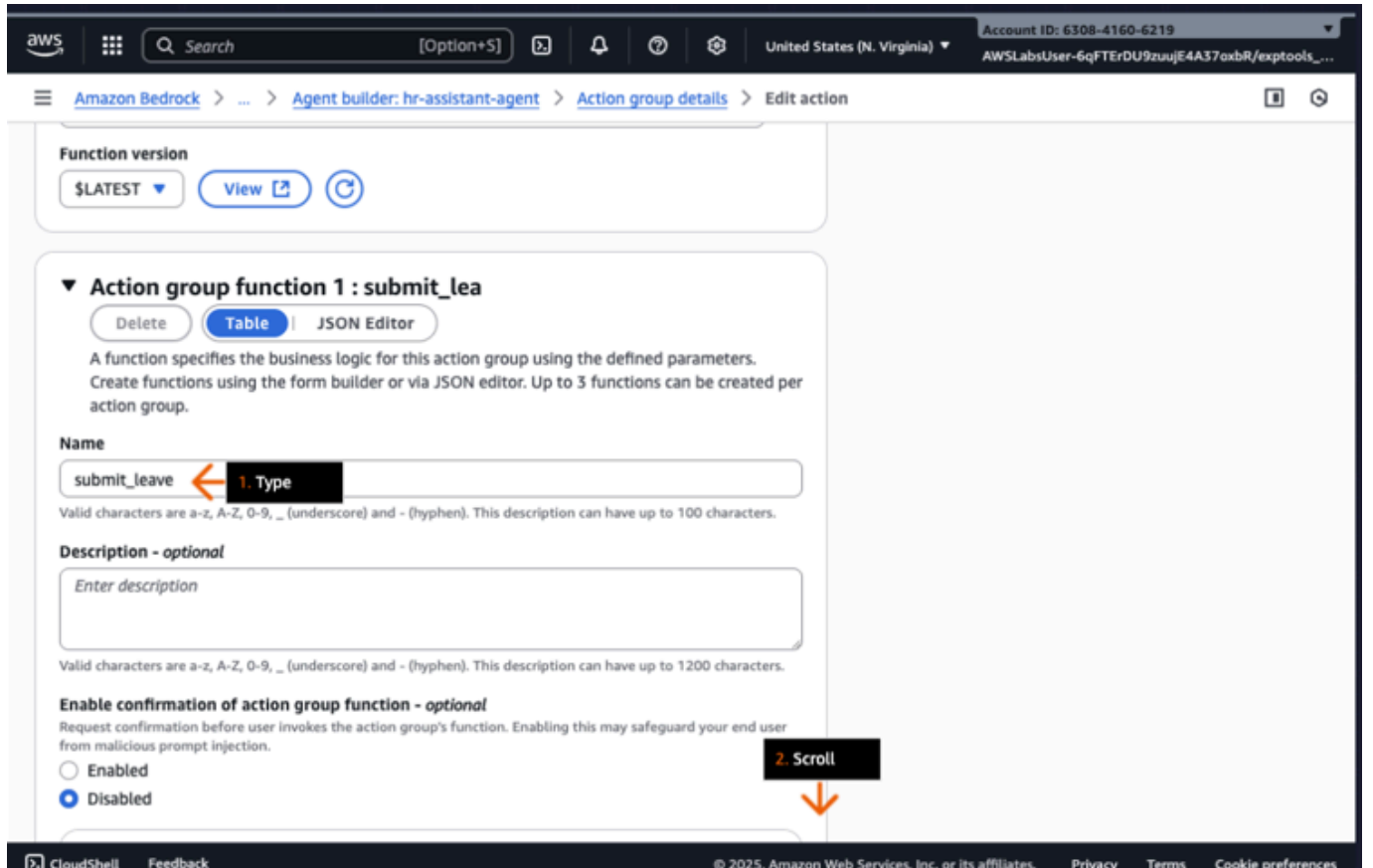
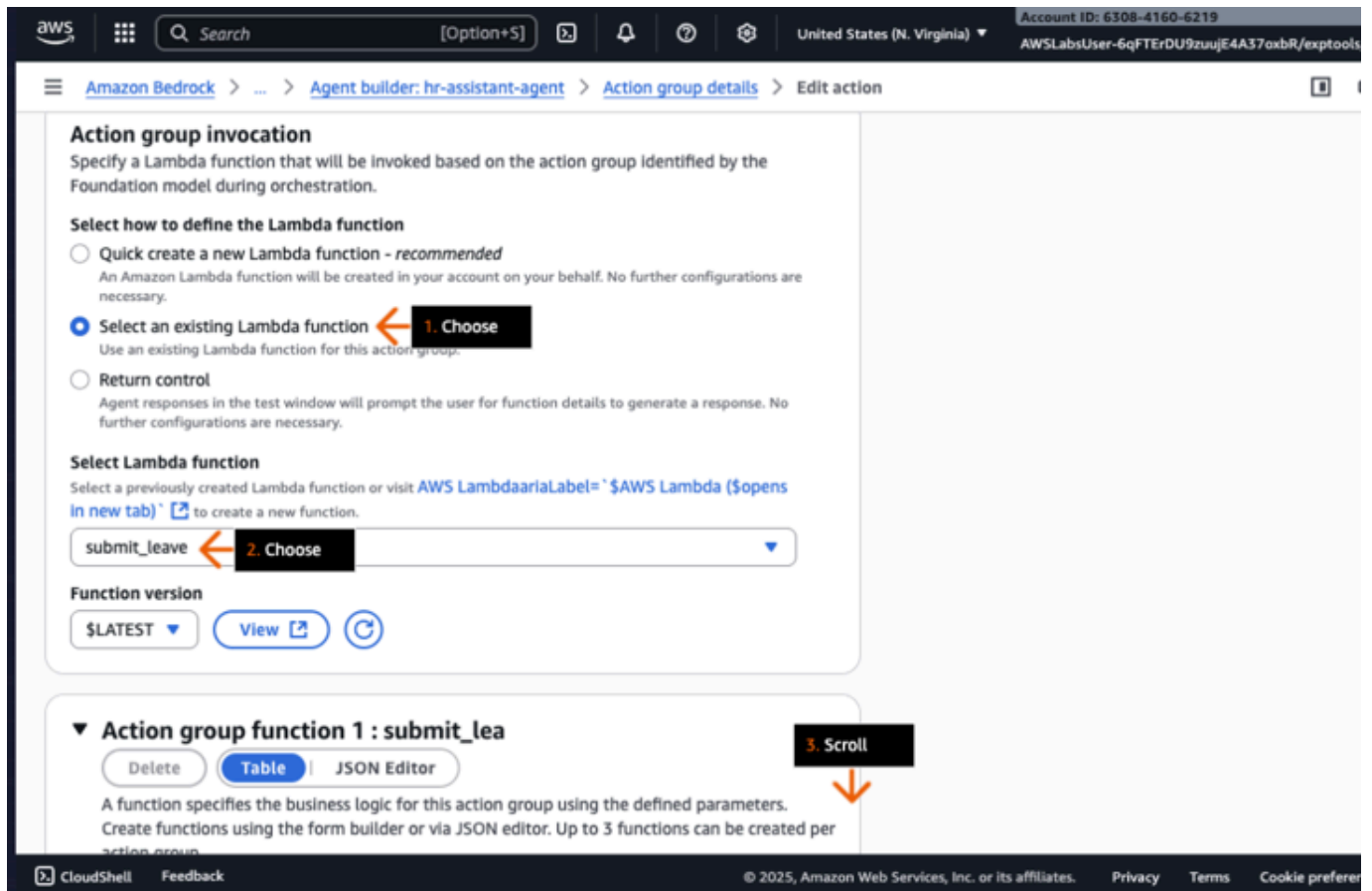
Permitirás que el agente "haga cosas", no solo que "hable".



1. **Añade un Action Group:** Dale el nombre `submit_leave_action`.



2. **Vincula Lambda:** Selecciona la función Lambda existente `submit_leave`.



3. Define Parámetros: Configura los parámetros que el agente debe extraer de la conversación:

- **employee_name** (String, Required)
- **startDate** (String, Required)
- **endDate** (String, Required)

The image consists of two screenshots of the Amazon Bedrock Agent Builder interface, specifically the 'Action group details' page for an agent named 'hr-assistant-agent'.

Top Screenshot: This screenshot shows the initial configuration of the action group. The 'Parameters' section contains a table with one parameter:

Name	Description	Type	Required	Delete
employee_name	Name of employee	string	True	

Annotations in this screenshot include:

- 1. Click**: Points to the 'Add action group function' button.
- 2. Type**: Points to the 'Name' column header.
- 3. Type**: Points to the 'Type' column header.
- 4. Choose**: Points to the 'Required' column header.

The 'Action status' section shows the 'Enable' radio button selected.

Bottom Screenshot: This screenshot shows the configuration after adding more parameters. The 'Parameters' table now has three rows:

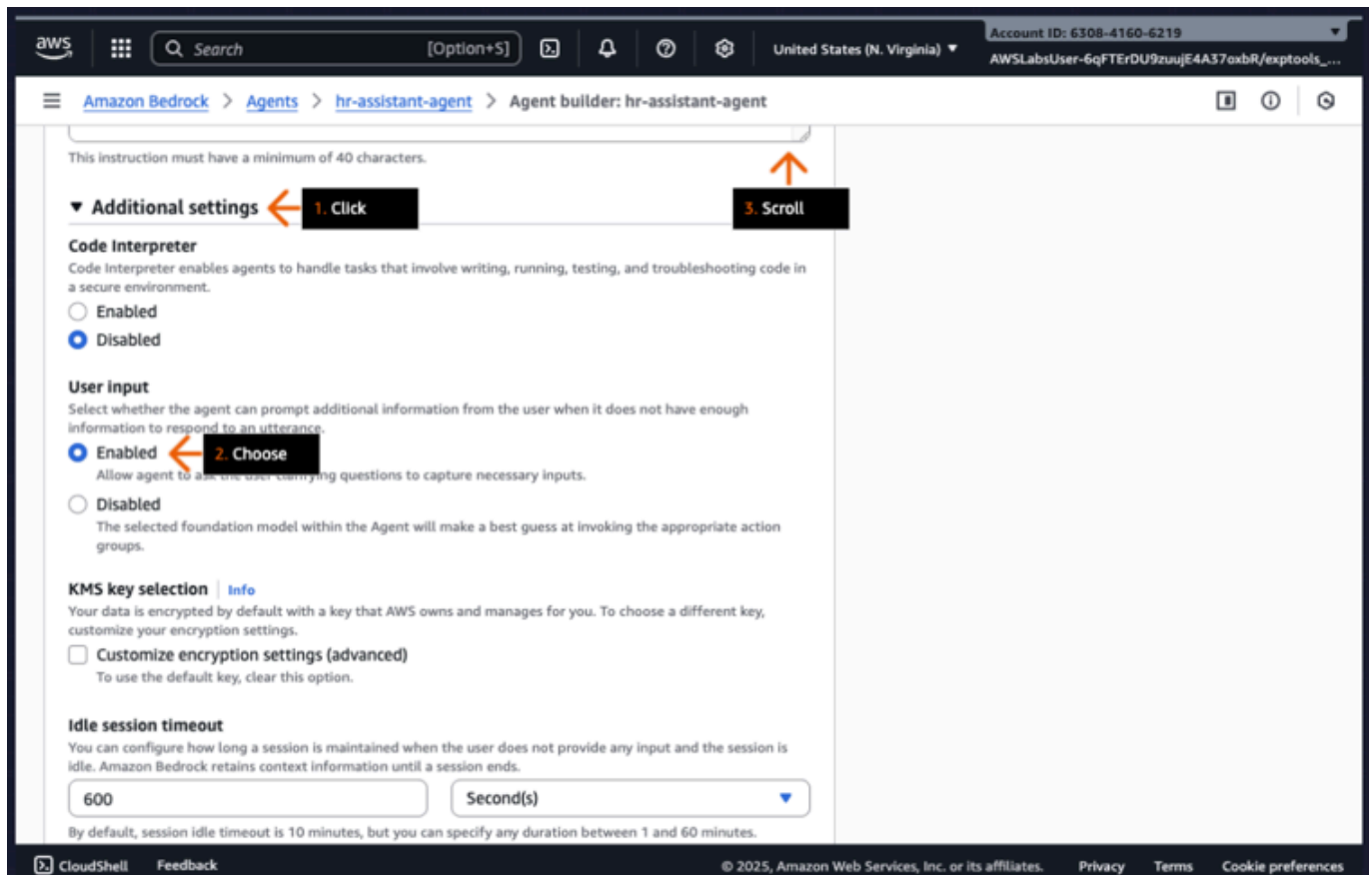
Name	Description	Type	Required	Delete
employee_name	Name of employee	string	True	
startDate	Start of leave	String	True	
endDate	End of leave	String	True	

Annotations in this screenshot include:

- 1. Click**: Points to the 'Add parameter' button.
- 2. Type**: Points to the 'Name' column header.
- 3. Review**: Points to the 'Required' column header.
- 4. Click**: Points to the 'Save and exit' button.

The 'Action status' section remains the same, with 'Enable' selected.

4. **Configura la Interacción:** Habilita el "User Input" para que el agente pueda preguntar detalles faltantes al usuario.



Tarea 6: Pruebas y Validación en DynamoDB

1. **Prepara el Agente:** Haz clic en **Save** y luego en **Prepare**.
2. **Prueba de Conocimiento:** En el chat de prueba, pregunta: "What is the vacation policy?". El agente debe responder citando los manuales de S3.
3. **Prueba de Acción:** Escribe: "submit my vacation request". El agente te pedirá tu nombre y fechas. Proporciona los datos (ej: Jane Doe, Jan 1 to June 30).
4. **Verificación Final:** Navega a **Amazon DynamoDB > Tables > VacationTable**. Haz clic en **Explore table items** y confirma que los datos de "Jane Doe" han sido insertados correctamente por el Agente.

aws | s3 | United States (N. Virginia) | AWSLabUser-93V5gsSRgAmDr8zJ9ghKiv/exptools_session @ 4931-9

Amazon Bedrock > Agents > hr-assistant-agent

Agent builder

Manual | Assistant | **Test** | Prepare | Save | Save and exit

Agent details

Agent name
hr-assistant-agent
Valid characters are a-z, A-Z, 0-9, _ (underscore) and - (hyphen). The name can have up to 100 characters.

Agent description - optional
Enter description
The description can have up to 200 characters.

Agent resource role
☐ Create and use a new service role
☒ Use an existing service role
arn:aws:iam::493195925326:role/Bedrock_Agent_Role

Select model
Amazon Nova Pro 1.0

Test Agent
Using ODT Chan

submit my vacation request

Run

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aws | Search | [Option+S] | United States (N. Virginia) | Account ID: 6308-4160-6219 | AWSLabUser-6qFTerDU9zuujE4A37oxbR/exptools_...

Amazon Bedrock > Agents > hr-assistant-agent > Agent builder: hr-assistant-agent

Agent builder

Manual | Assistant | **Test** | Prepare | Save | Save and exit

Agent details

Agent name
hr-assistant-agent
Valid characters are a-z, A-Z, 0-9, _ (underscore) and - (hyphen). The name can have up to 100 characters.

Agent description - optional
Enter description
The description can have up to 200 characters.

Agent resource role
☐ Create and use a new service role
☒ Use an existing service role
arn:aws:iam::630841606219:role/Bedrock_Agent_Role

Select model
Amazon Nova Pro 1.0

Test Agent
Using ODT Chan

Please provide your name.
Jane Doe
Show trace >

Please provide the start date of your leave.
Jan 1 2026

Run

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The screenshot shows the 'Agent builder: hr-assistant-agent' page in the Amazon Bedrock console. The left sidebar contains the 'Agent description - optional' field, 'Agent resource role' (set to 'Use an existing service role' with 'arn:aws:iam::630841606219:role/Bedrock_Agent_Role'), 'Select model' (set to 'Nova Pro 1.0'), and 'Instructions for the Agent' (set to 'You are a professional HR Assistant responsible for helping employees with HR-related policy questions. Follow these guidelines:'). The right sidebar shows the 'Test Agent' section with a 'Run' button. A red arrow labeled '1. Review' points to the 'Test Agent' section.

Account ID: 6308-4160-6219
AWS Labs User-6qFTErDU9zuujE4A37oxbR/exptools_...

United States (N. Virginia)

Amazon Bedrock > Agents > hr-assistant-agent > Agent builder: hr-assistant-agent

valid characters are a-z, A-Z, 0-9, _ (underscore) and - (hyphen). The name can have up to 100 characters.

Agent description - optional

Enter description

The description can have up to 200 characters.

Agent resource role

☐ Create and use a new service role

☒ Use an existing service role

arn:aws:iam::630841606219:role/Bedrock_Agent_Role

Select model

Nova Pro 1.0

Instructions for the Agent

Provide clear and specific instructions for the task the Agent will perform. You can also provide certain style and tone.

You are a professional HR Assistant responsible for helping employees with HR-related policy questions. Follow these guidelines:

This instruction must have a minimum of 40 characters.

Additional settings

Test Agent

Using ODT Chan

Please provide the end date of your leave.

June 30 2026

Your leave request from Jan 1, 2026, to June 30, 2026, has been successfully submitted. Thank you for using our

June 30 2026

Run

CloudShell Feedback

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The screenshot shows the Amazon Bedrock console with a search bar containing 'lambda'. The search results for 'lambda' are displayed, showing 'Services' and 'Features'. The 'Services' section lists 'Lambda', 'CodeBuild', and 'AWS Signer'. The 'Features' section lists 'Lambda Insights', 'Local processing', and 'Object Lambda Access Points'. A red arrow labeled '1. Type' points to the search bar, and another red arrow labeled '2. Click' points to the 'Lambda' service card.

Services

Features

Resources **New**

Documentation

Knowledge articles

Marketplace

Blog posts

Events

Tutorials

Search results for 'lambda'

Services Show more ▶

Lambda ← 2. Click
Run code without thinking about servers

CodeBuild ☆
Build and Test Code

AWS Signer ☆
Ensuring trust and integrity of your code

Features Show more ▶

Lambda Insights
CloudWatch feature

Local processing
IoT Core feature

Object Lambda Access Points
S3 feature

Account ID: d7AmAWvQs3Gom3P4Jv1pTZ/exptools_session @

United States (N. Virg)

Amazon Bedrock > Agent bui

Manual

Save and exit

Agent det

Agent name

hr-assistant

Valid characters characters.

Agent descrip

Enter descri

The description

Agent resourc

☐ Create and

☒ Use an exi

arn:aws:lam

Select model

Nova Pro 1.0

CloudShell Feedback

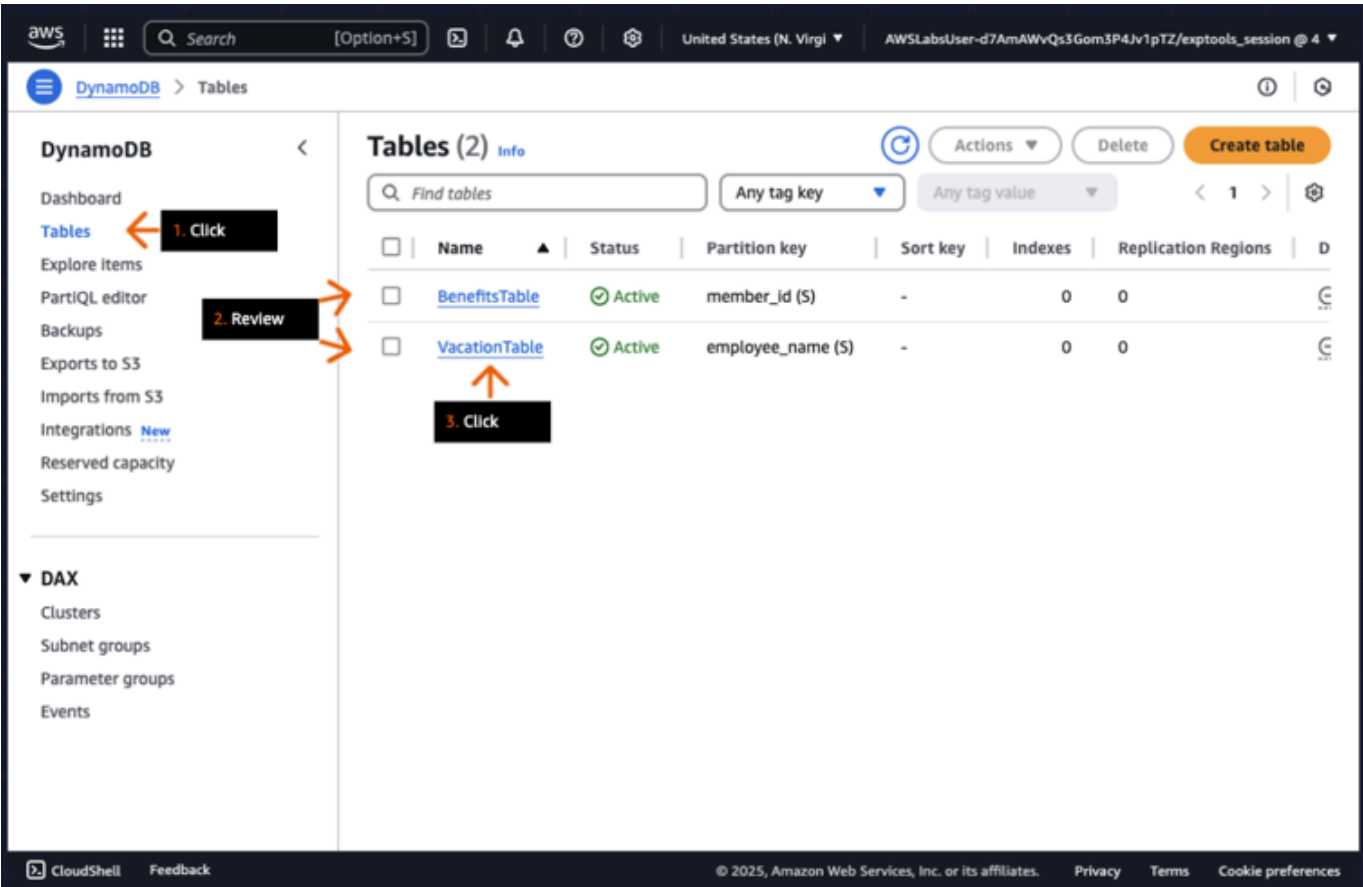
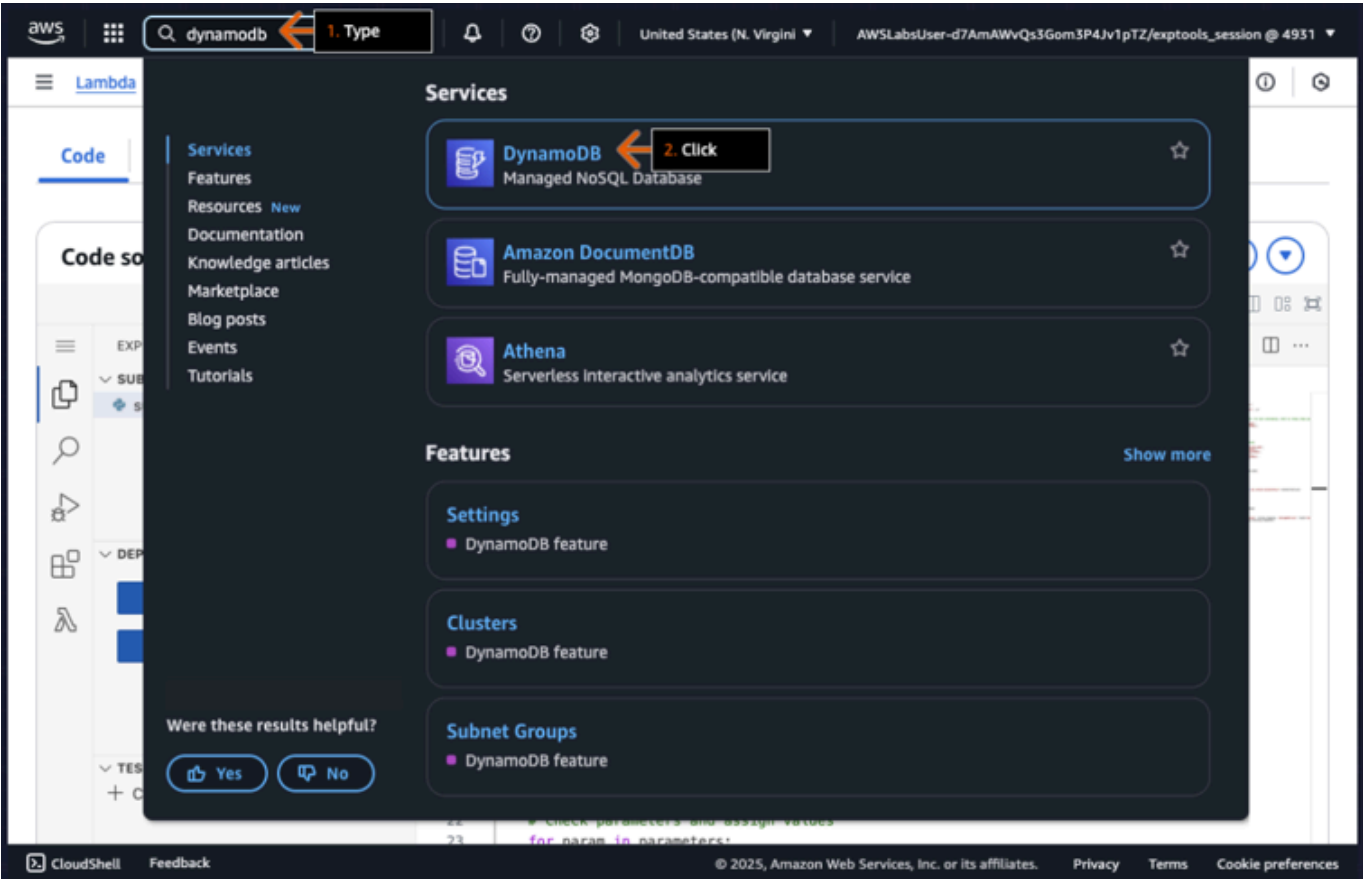
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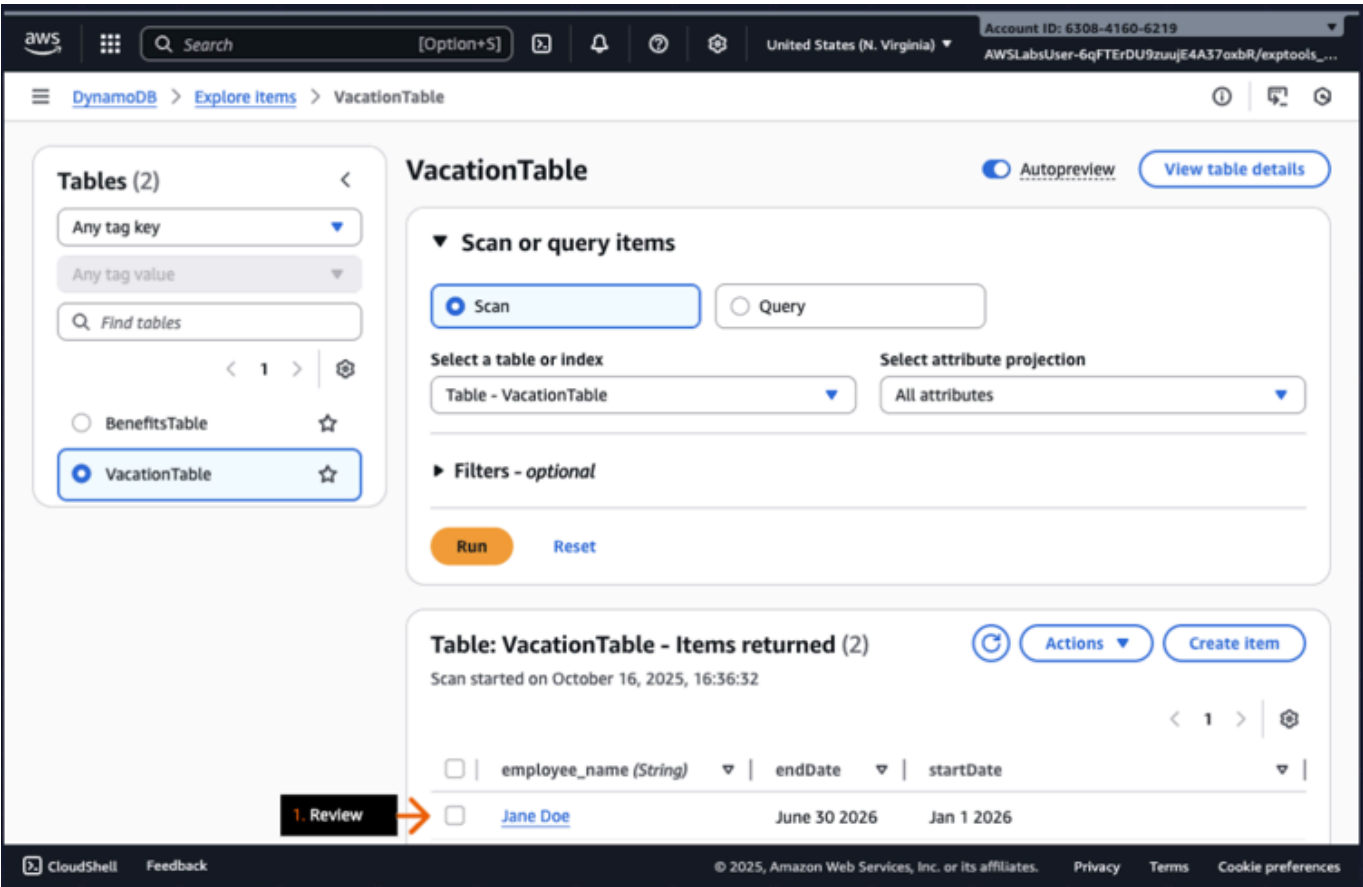
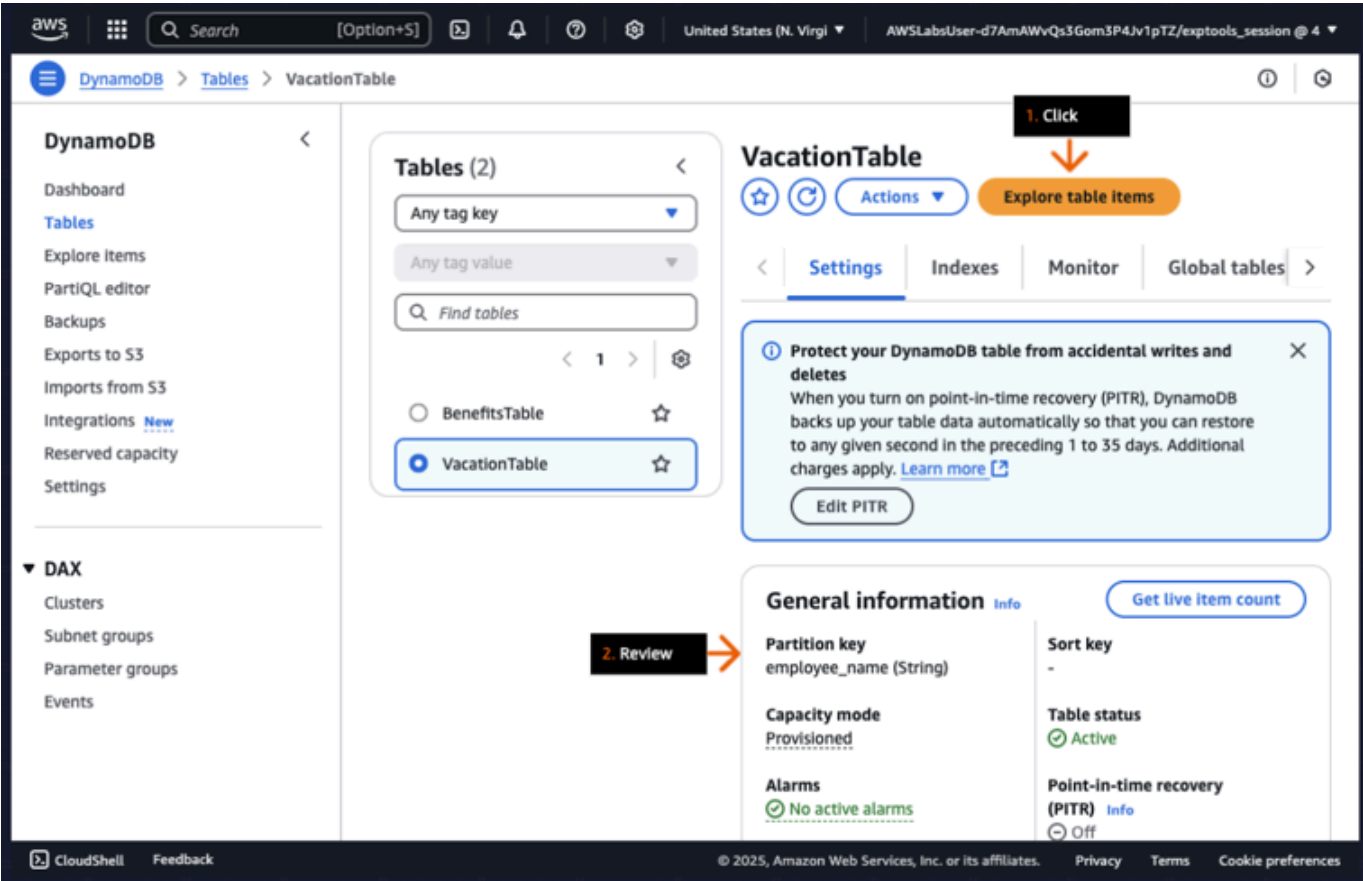
The screenshot shows the AWS Lambda console. On the left, there's a sidebar with 'Lambda' and 'Functions' links. Below that, there are sections for 'Additional resources' and 'Related AWS resources'. The main area displays a list of functions. The function 'submit_leave' is highlighted with a red arrow and a '2. Click' label. The function 'LabStack-22a374d6-0c5f-42-GblLabMonitoringgblprote-mYFxlW0ASf0Z' is also highlighted with a red arrow and a '1. Review' label.

Function name	Package type	Runtime
submit_benefits	Zip	Python 3.11
load_data_function	Zip	Python 3.11
LabStack-22a374d6-0c5f-42-IndexCreationHandlerA9D1-2p2HYIiot7J4	Zip	Python 3.11
LabStack-22a374d6-0c5f-42-GblLabMonitoringgblprote-mYFxlW0ASf0Z	Zip	Python 3.11
submit_leave	Zip	Python 3.13
gbl_lab_monitoring	Zip	Python 3.11

The screenshot shows the AWS Lambda console with the 'submit_leave' function selected. The 'Code' tab is active, displaying the code source for 'submitLeave.py'. The code is as follows:

```
1 import boto3
2
3
4 def lambda_handler(event, context):
5     agent = event['agent']
6     actionGroup = event['actionGroup']
7     function = event['function']
8     parameters = event.get('parameters', [])
9
10    print(parameters)
11
12    # Execute your business logic here. For more information, refer to: https://doc
13
14    dynamodb = boto3.resource('dynamodb')
15    table = dynamodb.Table('VacationTable')
16
17    # Initialize variables
18    employee_name = None
19    startDate = None
20    endDate = None
21
22    # Check parameters and assign values
23    for param in parameters:
```





5. Respuestas Analíticas

- **¿Cuál es la ventaja de usar Nova Pro en lugar de un modelo estándar?** El modelo **Nova Pro** está específicamente optimizado para la orquestación de agentes. Posee una mayor capacidad de

razonamiento para determinar cuándo debe consultar la base de conocimientos y cuándo debe invocar una función Lambda (Tool Use/Function Calling) con alta precisión en la extracción de parámetros.

- **¿Por qué es necesario el proceso de "Sync" en la Knowledge Base?** El proceso de sincronización realiza el *chunking* (división del texto) y la vectorización de los documentos. Sin este paso, el modelo de lenguaje no podría realizar búsquedas semánticas sobre los datos de S3, ya que los LLM no leen archivos planos en tiempo de ejecución, sino que consultan representaciones matemáticas (vectores) en OpenSearch.
- **¿Qué función cumple Lambda en este ecosistema?** Lambda actúa como el ejecutor de efectos secundarios. Mientras que el Agente procesa el lenguaje natural, Lambda traduce las intenciones del Agente en llamadas de API o consultas de base de datos (en este caso, un **PutItem** en DynamoDB), permitiendo que la IA interactúe con sistemas transaccionales legacy o modernos.

Documentación generada por Alejandro Barrenechea.

April, 2026
